

The Optimizer Quotient and the Certification Trilemma

Tristan Simas

Abstract

The optimizer quotient is the canonical object for exact decision-relevant information: it is the coarsest exact decision-preserving abstraction (Theorem 2.15). This paper proves that exact certification of this object’s coordinate structure is subject to an impossibility trilemma: under $P \neq \text{coNP}$, no certifier can be simultaneously sound, complete on all in-scope instances, and polynomial-budgeted (Theorem 7.1). The cost of this impossibility varies by regime: coNP (static), PP-hard (stochastic decisiveness), PSPACE-complete (sequential). Six structural restrictions collapse certification to polynomial time. The finite reduction and verification core is mechanized in Lean 4.

I. INTRODUCTION

A. *The Optimizer Quotient: A Canonical Object*

Exact decision-preserving information has a canonical form: the optimizer quotient. For a finite coordinate-structured decision problem $\mathcal{D} = (A, S, U)$ with $S = X_1 \times \cdots \times X_n$, the optimizer quotient identifies states exactly when they induce the same optimal-action set $\text{Opt}(s)$. This quotient is the coarsest exact decision-preserving abstraction (Theorem II.15): every surjective abstraction that preserves optimal actions factors uniquely through it. Equivalently, in **Set**, it is the coimage/image factorization of the optimizer map.

This canonical object exists for every decision problem and unifies separate literatures. In rough-set theory, attribute reduction asks which coordinates preserve decision distinctions; in constraint satisfaction, backdoor sets identify variables whose fixation yields tractability; in feature selection, minimal sufficient subsets are sought; in MDP abstraction, state aggregation preserves value or policy structure. The optimizer quotient provides a common mathematical foundation: it is the canonical object for exact relevance in all these settings.

B. The Certification Trilemma: An Impossibility Theorem

From the optimizer quotient we derive a structural constraint on exact certifiers. Under $P \neq \text{coNP}$, no solver can be simultaneously integrity-preserving, non-abstaining on all in-scope instances, and polynomial-budgeted (Theorem 7.1). This impossibility yields a trilemma: any exact certifier in the hard regime must either abstain on some instances, weaken its guarantee, or risk overclaiming. The trilemma is not merely a restatement of coNP-hardness; it is a certified-solver impossibility result with operational consequences for systems claiming exact reliability.

C. The Complexity Map as Evidence

The impossibility manifests differently across regimes. The static regime yields coNP-completeness for base and minimum queries (via relevance containment), while static anchor sufficiency is Σ_2^P -complete. In the stochastic regime, the problem splits: preservation bridges to static sufficiency (polynomial-time under explicit encoding), while decisiveness is PP-hard under succinct encoding. In the sequential regime, temporal contingency lifts all queries to PSPACE-complete. These classifications form a regime-sensitive complexity matrix (Table III) that shows the impossibility’s reach expanding with model expressiveness. Two narrow cells in the stochastic-preservation row (general distributions, without the full-support assumption) remain open; these are explicitly marked as scope boundaries in Table II and do not affect any of the proved results, the optimizer quotient, the certification trilemma, or the completeness of the static, sequential, and stochastic-decisiveness regimes.

D. Tractable Boundaries

Six structural restrictions collapse exact certification to polynomial time: bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, and coordinate symmetry. Each removes a distinct source of hardness, providing tractable boundaries to the impossibility theorem.

E. Mechanized Core

A Lean 4 artifact mechanically checks the optimizer-quotient universal property, finite deciders, reduction-correctness lemmas, witness schemas, and bridge results¹ The 22K-line artifact verifies the combinatorial core, while oracle-class placements are argued in the paper text.

¹ Lean: QT1, QT7, DC91–99, OU1–2, OU8–12, EH1–10.

F. Paper Structure

Section II introduces the formal setup. Sections III, IV, and V develop the static, stochastic, and sequential complexity classifications. Section VI consolidates them into the regime matrix. Section 7 presents the impossibility theorem and trilemma. Section 8 collects structural consequences. Sections VII and VIII cover encoding contrasts and tractable cases. Section 9 situates the work, and Appendix A collects model translations.

G. Artifact Availability

The proof artifact is archived at <https://doi.org/10.5281/zenodo.19057595>. The cited-content snapshot contains 22068 lines across 67 files with 0 occurrences of `sorry`.

II. FORMAL SETUP

This section fixes the abstract vocabulary used throughout the paper. We work with finite decision problems whose state spaces factor into coordinates, and we isolate the minimal terminology needed to state the later complexity results. The same setup will support the paper’s three core regime classifications, the encoding-sensitive boundary, and the later consequence sections.

Object	Quantifier shape	Certifies	Role
SUFFICIENCY-CHECK	universal over state pairs	exact coordinate sufficiency	baseline certification predicate
MINIMUM-SUFFICIENT-SET	apparent $\exists I \forall (s, s')$ check	smallest sufficient support	collapses to relevance containment in static regime
ANCHOR-SUFFICIENCY	$\exists$ anchor, then universal fiber check	exact preservation on one chosen fiber	retains the outer existential layer
Optimizer quotient	universal factorization property	coarsest decision-preserving abstraction	canonical structural object

TABLE I: Setup map. The four central objects differ mainly in quantifier shape: minimum sufficiency collapses in the static regime, while anchor sufficiency keeps its existential fiber choice.

A. Decision Problems with Coordinate Structure

Definition II.1 (Decision Problem). A *decision problem with coordinate structure* is a tuple $\mathcal{D} = (A, X_1, \dots, X_n, U)$ where:

- A is a finite set of *actions* (alternatives)

- $X_1, \dots, X_n$ are finite *coordinate spaces*
- $S = X_1 \times \dots \times X_n$ is the *state space*
- $U : A \times S \rightarrow \mathbb{Q}$ is the *utility function*

Definition II.2 (Projection). For state $s = (s_1, \dots, s_n) \in S$ and coordinate set $I \subseteq \{1, \dots, n\}$:

$$s_I := (s_i)_{i \in I}$$

is the *projection* of s onto coordinates in I .

Definition II.3 (Optimizer Map). For state $s \in S$, the *optimal action set* is:

$$\text{Opt}(s) := \arg \max_{a \in A} U(a, s) = \{a \in A : U(a, s) = \max_{a' \in A} U(a', s)\}$$

B. Sufficiency and Relevance

Definition II.4 (Sufficient Coordinate Set). A coordinate set $I \subseteq \{1, \dots, n\}$ is *sufficient* for decision problem $\mathcal{D}$ if:

$$\forall s, s' \in S : \quad s_I = s'_I \implies \text{Opt}(s) = \text{Opt}(s')$$

Equivalently, the optimal action depends only on coordinates in I .

Proposition II.5² (Empty-Set Sufficiency Equals Constant Decision Boundary). *The empty set $\emptyset$ is sufficient if and only if $\text{Opt}(s)$ is constant over the entire state space.*

Proof. If $\emptyset$ is sufficient, then every two states agree on their empty projection, so sufficiency forces $\text{Opt}(s) = \text{Opt}(s')$ for all $s, s' \in S$. The converse is immediate. ■

Proposition II.6³ (Insufficiency Equals Counterexample Witness). *Coordinate set I is not sufficient if and only if there exist states $s, s' \in S$ such that $s_I = s'_I$ but $\text{Opt}(s) \neq \text{Opt}(s')$.*

Proof. This is the direct negation of Definition II.4. ■

Definition II.7 (Minimal Sufficient Set). A sufficient set I is *minimal* if no proper subset $I' \subsetneq I$ is sufficient.

Definition II.8 (Relevant Coordinate). Coordinate i is *relevant* if there exist states that differ only at coordinate i and induce different optimal-action sets:

$$i \text{ is relevant} \iff \exists s, s' \in S : \left(\forall j \neq i, s_j = s'_j \right) \wedge \text{Opt}(s) \neq \text{Opt}(s').$$

² : DP6 ³ : DP7

Proposition II.9⁴ (Minimal-Set/Relevance Equivalence). *For any minimal sufficient set I and any coordinate i ,*

$$i \in I \iff i \text{ is relevant.}$$

Hence every minimal sufficient set is exactly the relevant-coordinate set.

Proof. If $i \in I$ were not relevant, then changing coordinate i while keeping all others fixed would never alter the optimal-action set, so removing i would preserve sufficiency, contradicting minimality. Conversely, every sufficient set must contain each relevant coordinate, because omitting a relevant coordinate leaves a witness pair with identical I -projection but different optimal-action sets. ■

Definition II.10 (Exact Relevance Identifiability). For a decision problem $\mathcal{D}$ and candidate set I , we say that I is *exactly relevance-identifying* if

$$\forall i \in \{1, \dots, n\} : \quad i \in I \iff i \text{ is relevant for } \mathcal{D}.$$

Definition II.11 (Structural Rank). The *structural rank* of a decision problem is the number of relevant coordinates:

$$\text{srnk}(\mathcal{D}) := |\{i \in \{1, \dots, n\} : i \text{ is relevant}\}|.$$

C. Optimizer Quotient

Definition II.12 (Decision Equivalence). States $s, s' \in S$ are *decision-equivalent* if they induce the same optimal-action set:

$$s \sim_{\text{Opt}} s' \iff \text{Opt}(s) = \text{Opt}(s').$$

Definition II.13 (Decision Quotient). The *decision quotient* (equivalently, optimizer quotient) of $\mathcal{D}$ is the quotient object

$$Q_{\mathcal{D}} := S / \sim_{\text{Opt}}.$$

Its equivalence classes are exactly the maximal subsets of states on which the optimal-action set is constant.

The decision quotient is the canonical mathematical object associated with exact decision preservation. It quotients the state space by optimizer-equivalence and thereby isolates precisely the distinctions that any decision-preserving abstraction must respect.

⁴ : DP1-2

Proposition II.14⁵ (Optimizer Quotient as Coimage/Image Factorization). *The decision quotient $Q_{\mathcal{D}}$ is canonically in bijection with $\text{range}(\text{Opt})$. Equivalently, in **Set**, it is the coimage/image factorization of the optimizer map.*

Proof. Two states are identified by $\sim_{\text{Opt}}$ exactly when they have the same optimizer value. Hence quotient classes are in one-to-one correspondence with attained values of Opt . ■

Theorem II.15⁶ (Universal Property of the Optimizer Quotient). *Let $Q_{\mathcal{D}} = S/\sim_{\text{Opt}}$ be the decision quotient with quotient map $\pi : S \rightarrow Q_{\mathcal{D}}$. For any surjective abstraction $\phi : S \rightarrow T$ such that*

$$\phi(s) = \phi(s') \implies \text{Opt}(s) = \text{Opt}(s'),$$

there exists a unique map $\psi : T \rightarrow Q_{\mathcal{D}}$ such that

$$\pi = \psi \circ \phi.$$

Proof. Because ϕ preserves optimizer values on its fibers, any two states identified by ϕ are decision-equivalent. So each fiber of ϕ is contained in a single $\sim_{\text{Opt}}$ -class.

For each $t \in T$, choose any $s \in S$ with $\phi(s) = t$ and define $\psi(t) := \pi(s)$. This is well defined because any two such choices lie in the same $\sim_{\text{Opt}}$ -class. By construction, $(\psi \circ \phi)(s) = \pi(s)$ for every $s \in S$, so $\pi = \psi \circ \phi$. Uniqueness follows from surjectivity of ϕ : if ψ' also satisfies $\pi = \psi' \circ \phi$, then every $t \in T$ has the form $t = \phi(s)$ and therefore

$$\psi(t) = \psi(\phi(s)) = \pi(s) = \psi'(\phi(s)) = \psi'(t).$$

Hence $\psi = \psi'$. ■

Remark II.16⁷ (Canonical Status). The optimizer quotient is therefore the coarsest abstraction of the state space that preserves optimal-action distinctions. Any surjective abstraction that is strictly coarser must erase some decision-relevant distinction.

Remark II.17⁸ (Decision Noise). The quotient viewpoint identifies decision noise exactly. At the state level, decision noise is the kernel of the optimizer map: two states differ only by decision noise if and only if they determine the same class in $Q_{\mathcal{D}}$. At the coordinate level, a coordinate is pure decision noise if and only if varying it alone never changes the decision, equivalently if and only if it is irrelevant. In the

⁵ : DQ6-7 ⁶ : QT1, QT7 ⁷ : QT7, AB2. ⁸ : DN1-2, DN5.

finite probabilistic reading used later, this is also equivalent to conditional independence of the decision from that coordinate given the remaining coordinates.

Proposition II.18⁹ (Sufficiency Characterization). *Coordinate set I is sufficient if and only if the optimizer map factors through the projection $\pi_I : S \rightarrow S_I$. Equivalently, there exists a function $\overline{\text{Opt}}_I : S_I \rightarrow \mathcal{P}(A)$ such that*

$$\text{Opt} = \overline{\text{Opt}}_I \circ \pi_I.$$

Proof. If I is sufficient, define $\overline{\text{Opt}}_I(x)$ to be $\text{Opt}(s)$ for any state s with $s_I = x$; this is well defined exactly because sufficiency makes Opt constant on each projection fiber. Conversely, any such factorization forces states with equal I -projection to have equal optimal-action sets. ■

D. Computational Model and Input Encoding

The same decision question yields different computational problems under different input models. We therefore fix the encoding regime before stating any complexity classification.

a) Succinct encoding (primary for hardness).: An instance is encoded by:

- a finite action set A given explicitly,
- coordinate domains $X_1, \dots, X_n$ given by their sizes in binary,
- a Boolean or arithmetic circuit C_U that on input (a, s) outputs $U(a, s)$.

The input length is

$$L = |A| + \sum_i \log |X_i| + |C_U|.$$

Unless stated otherwise, all class-theoretic hardness results in this paper are measured in L .

b) Explicit-state encoding.: The utility function is given as a full table over $A \times S$. The input length is

$$L_{\text{exp}} = \Theta(|A||S|)$$

up to the bitlength of the utility entries. Polynomial-time claims stated in terms of $|S|$ use this explicit-state regime.

c) Query-access regime.: The solver is given oracle access to $\text{Opt}(s)$ or to the utility values needed to reconstruct it. Complexity is then measured by query count, optionally paired with per-query evaluation

⁹ : [DQ42](#)

cost. This regime separates access obstruction from full-table availability and from succinct circuit input length.

Remark II.19 (Decision Questions vs Decision Problems). The predicate “is coordinate set I sufficient for $\mathcal{D}$?” is an encoding-independent mathematical question. A *decision problem* arises only after fixing how $\mathcal{D}$ and I are represented. Complexity classes are properties of these encoded problems, not of the abstract predicate in isolation.

Remark II.20 (Later Stochastic Comparison). In the stochastic row, the preservation predicate compares the optimizer seen under coarse conditioning on I with the fully conditioned optimizer. Under full support, the later bridge package reconnects this stochastic notion to the same quotient picture as the static regime.

III. STATIC REGIME COMPLEXITY

This section studies the static regime, where the input is a finite coordinate-structured decision problem under the succinct encoding of Section II-D. This is the baseline certification setting: no stochastic averaging, no temporal dynamics, only the exact question of whether a candidate coordinate set preserves the optimizer map. The main contribution here is structural: sufficiency is exactly relevance containment. Once that characterization is in place, the complexity classification for MINIMUM-SUFFICIENT-SET follows immediately. The only genuinely higher-level static query is the anchor variant, where an existentially chosen fiber must satisfy a universal constancy condition.

A. Problem Definitions

Problem III.1 (SUFFICIENCY-CHECK). **Input:** Decision problem $\mathcal{D} = (A, X_1, \dots, X_n, U)$ and coordinate set $I \subseteq \{1, \dots, n\}$.

Question: Is I sufficient for $\mathcal{D}$?

Problem III.2 (MINIMUM-SUFFICIENT-SET). **Input:** Decision problem $\mathcal{D} = (A, X_1, \dots, X_n, U)$ and integer k .

Question: Does there exist a sufficient set I with $|I| \leq k$?

Problem III.3 (ANCHOR-SUFFICIENCY). **Input:** Decision problem $\mathcal{D} = (A, X_1, \dots, X_n, U)$ and fixed coordinate set I .

Question: Does there exist an assignment α to I such that $\text{Opt}(s)$ is constant on the fiber $\{s : s_I = \alpha\}$?

B. Apparent Σ_2^P Structure

At first glance, MINIMUM-SUFFICIENT-SET seems to have the form

$$\exists I \forall s, s' \in S : s_I = s'_I \implies \text{Opt}(s) = \text{Opt}(s').$$

That syntax suggests a Σ_2^P classification. The algorithm suggested by this structure is: guess a coordinate set I , then universally verify that every agreeing state pair induces the same optimal-action set. On its face, static exact minimization therefore looks like a genuine existential–universal search problem.

C. Structural Characterization of Minimum Sufficiency

Proposition II.9 removes the outer search layer: a set is sufficient exactly when it contains every relevant coordinate, so the minimum sufficient set is simply the relevant-coordinate set. This is the central structural theorem of the static regime. Static exact minimization is therefore a relevance-counting problem, not a genuinely alternating search problem.

That collapse is specific to minimum sufficiency. ANCHOR-SUFFICIENCY still asks the algorithm to choose an assignment α and then certify a universal condition on the induced fiber, so its existential layer cannot be absorbed into a structural characterization of the instance.

D. Complexity Consequence: coNP-Completeness

The first two results show that exact certification is already harder than pointwise evaluation. A counterexample to sufficiency is short, but certifying that no such counterexample exists is a universal statement over state pairs. The collapse discussed above explains why the same coNP behavior governs both direct sufficiency checking and exact minimization.

a) Proof idea.: Membership is immediate from short counterexamples: a “no” witness is just a pair of states that agree on I but induce different optimal-action sets. Hardness comes from reducing TAUTOLOGY to an empty-information instance whose optimizer map is constant exactly in the tautology case.

Theorem III.4¹⁰ (SUFFICIENCY-CHECK is coNP-complete). *SUFFICIENCY-CHECK is coNP-complete.*

Proof. For membership, Proposition II.6 gives a polynomial witness (s, s') with $s_I = s'_I$ and $\text{Opt}(s) \neq \text{Opt}(s')$ whenever I is not sufficient, so the complement is in NP.

¹⁰ : DQ40–41

For hardness, reduce TAUTOLOGY to the empty-set instance. Given formula $\varphi(x_1, \dots, x_n)$, build a two-action decision problem where one action tracks φ and the other is a baseline action. Then $I = \emptyset$ is sufficient iff φ is a tautology, because empty-set sufficiency is exactly constancy of Opt (Proposition II.5). The reduction is polynomial in formula size. ■

Theorem III.5¹¹ (Static Collapse Theorem). *In the static regime, the apparent existential layer in MINIMUM-SUFFICIENT-SET collapses to relevance containment. Consequently, MINIMUM-SUFFICIENT-SET is coNP-complete.*

b) Proof idea.: Once sufficiency is equivalent to containing all relevant coordinates, the question “is there a sufficient set of size at most k ?” becomes “are there at most k relevant coordinates?” The search component disappears, and the problem inherits the same coNP character as direct sufficiency checking.

Proof. Proposition II.9 identifies the minimum sufficient set with the relevant-coordinate set. The question “ $|I^*| \leq k$?” therefore becomes the question whether at most k coordinates are relevant, which is a coNP check over relevance witnesses. CoNP-hardness follows from the $k = 0$ slice, which is exactly SUFFICIENCY-CHECK with $I = \emptyset$. ■

E. Sigma-2-P Completeness

Anchor sufficiency adds one more layer of choice that the previous collapse does not remove: the algorithm may select a fiber, but must then certify that the chosen fiber is uniformly decision-preserving. That existential-then-universal pattern is exactly what raises the complexity class.

Theorem III.6¹² (ANCHOR-SUFFICIENCY is Sigma-2-P-complete). *ANCHOR-SUFFICIENCY is Σ_2^P -complete.*

Proof. Membership follows directly from the quantifier pattern:

$$\exists \alpha \forall s \in S : (s_I = \alpha) \Rightarrow \text{Opt}(s) = \text{Opt}(s_\alpha).$$

For hardness, reduce from $\exists\forall$ -SAT: existential variables encode the anchor assignment, and universal variables range over the residual coordinates. Utility values are set so that fiberwise constancy holds exactly when the source formula is true. ■

¹¹ : DQ35–36 ¹² : DQ98, CC4

F. Certificate-Size Lower Bound

The next theorem sharpens the static picture. It is not only that exact certification is hard in the complexity-class sense; even a checker restricted to the empty-set core may be forced to inspect exponentially many witness locations in the worst case.

Theorem III.7¹³ (Witness-Checking Duality). *For Boolean state spaces $S = \{0, 1\}^n$, any sound checker for the empty-set sufficiency core must inspect at least 2^{n-1} witness pairs in the worst case.*

Proof. Empty-set sufficiency asks whether Opt is constant on all 2^n states. A sound refutation-complete checker must be able to separate constant maps from maps that differ on some hidden antipodal partition. An adversary can place the first disagreement in any of 2^{n-1} independent pair slots; if fewer than 2^{n-1} slots are inspected, two instances remain indistinguishable to the checker but have opposite truth values. Therefore at least 2^{n-1} pair checks are necessary. ■

a) Mechanization note.: The TAUTOLOGY reduction stack, the coNP and Σ_2^P classifications, and the witness-budget lower-bound core are indexed by their inline Lean handles.

b) Explicit-state search upper bounds.: Under the explicit-state step-counting model, static sufficiency and static anchor sufficiency are also wrapped as abstract P predicates on inputs carrying a certified state budget. The artifact proves these upper bounds via counted searches with quadratic and linear state-space dependence respectively, and it also packages the static sufficiency search as an explicit correctness-plus-step-bound witness and as part of the unified finite-search summary theorem.¹⁴

This completes the baseline regime. The next section keeps the same exact certification question but replaces pairwise counterexample exclusion by conditional expected-utility comparison, which is why the complexity lifts from the static coNP/ Σ_2^P picture to PP.

IV. STOCHASTIC REGIME COMPLEXITY

We now add a distribution over states. Once utilities are compared through conditional expectations, the stochastic regime has two exact predicates rather than one. The first is a preservation predicate: does the optimizer seen after hiding coordinates agree with the optimizer seen when the full state is known? The second is a decisiveness predicate: does each observed fiber admit a unique Bayes-optimal action? Formally, these are stochastic sufficiency and stochastic decisiveness. Stochastic sufficiency gives the direct stochastic analogue of static sufficiency and the bridge back to the optimizer quotient; under full support,

¹³ : WD1-3 ¹⁴ : DC70-71, DC76, DC80-82.

its minimum and anchor variants inherit the static coNP and Σ_2^P classifications. Stochastic decisiveness yields the strongest succinct-encoding complexity classification established in the paper.

A. Preservation and Decisiveness

In static decision problems, the exact question is whether hiding coordinates preserves the optimizer. In stochastic decision problems, preservation is still a well-defined exact question, but it is no longer the only one. One can also ask whether the retained signal is itself a complete decision interface, meaning that every observable fiber has a unique conditional optimum. That is the decisiveness predicate. The two predicates are not equivalent: a coordinate set may preserve the full-information optimizer without being decisive, and it may be decisive without preserving the full-information optimizer. Probability therefore produces two parallel exact questions rather than one uniform stochastic analogue.

Example IV.1 (Preservation without Decisiveness). Consider a decision problem with two coordinates $X_1, X_2 \in \{0, 1\}$, actions $A = \{0, 1\}$, uniform distribution over states, and utility $U(a, s) = 1$ if $a = s_1 \oplus s_2$ (XOR), 0 otherwise. Hiding X_2 preserves the optimizer: given any value of X_1 , the conditional optimal action is still $s_1 \oplus s_2$ (since the distribution is uniform, the conditional expectation equals pointwise utility). However, X_1 alone is not decisive: for each fixed X_1 , both $s_2 = 0$ and $s_2 = 1$ yield different optimal actions (0 and 1, or 1 and 0), so no single action is conditionally optimal across the entire fiber.

Example IV.2 (Decisiveness without Preservation). Consider the same coordinate structure but with utility $U(a, s) = 1$ if $a = s_1$, 0 otherwise. Here X_1 is decisive: for each x_1 , action x_1 is uniquely optimal. However, X_1 does not preserve the full optimizer: when s_2 is revealed, the optimal action depends on s_1 alone, but the utility values differ across s_2 , so the conditional expectation at $X_1 = x_1$ averages over utilities that X_1 cannot distinguish.

B. Stochastic Decision Problems

Definition IV.3 (Stochastic Decision Problem). A stochastic decision problem is a tuple

$$\mathcal{D}_S = (A, X_1, \dots, X_n, U, P),$$

where $P \in \Delta(S)$ is a distribution on $S = X_1 \times \dots \times X_n$.

Definition IV.4 (Distribution-Conditional Sufficiency). For each admissible fiber value $\alpha \in \text{supp}(S_I)$, define the conditional optimal-action set

$$\text{Opt}_I^{\text{stoch}}(\alpha) := \arg \max_{a \in A} \mathbb{E}[U(a, S) \mid S_I = \alpha].$$

A coordinate set I is *stochastically sufficient* if conditioning on I preserves the fully conditioned optimizer:

$$\forall s \in S : \quad \text{Opt}_I^{\text{stoch}}(s_I) = \text{Opt}(s).$$

Here $S \sim P$ is the random state drawn from the instance distribution.

a) Intuition.: The conditional optimizer $\text{Opt}_I^{\text{stoch}}(\alpha)$ aggregates expected utilities over the entire fiber $\{s : s_I = \alpha\}$ using the distribution P . By contrast, the full-information optimizer $\text{Opt}(s)$ conditions on the exact state s itself. Preservation asks whether this averaging over a fiber ever changes which action is optimal; decisiveness asks whether, after averaging, the optimal action on each fiber is uniquely determined.

Definition IV.5 (Stochastic Decisiveness). Using the same conditional optimizer map $\text{Opt}_I^{\text{stoch}}$, a coordinate set I is *stochastically decisive* if every admissible fiber has a uniquely determined conditional optimal action:

$$\forall \alpha \in \text{supp}(S_I) \exists a \in A : \quad \text{Opt}_I^{\text{stoch}}(\alpha) = \{a\}.$$

In the paper, we refer to this as *stochastic decisiveness*. In the current artifact, this is the stochastic predicate with the strongest succinct-encoding complexity package.

b) From witness exclusion to counting.: Static sufficiency asks whether any counterexample pair exists. Stochastic sufficiency asks whether the conditional optimizer obtained from the retained coordinates still matches the fully conditioned optimizer, while stochastic decisiveness asks whether every admissible fiber has a uniquely determined conditional optimum. The first question is about ruling out disagreeing state pairs; the second and third are about conditional expected-utility comparisons on each fiber. That shift from witness exclusion to weighted counting comparison is exactly what brings stochastic hardness into the picture.

c) Example.: Consider a stochastic decision problem with $S = \{s_1, s_2\}$, $A = \{a, b\}$, uniform distribution $P(s_1) = P(s_2) = 1/2$, and utilities $U(a, s_1) = 2$, $U(b, s_1) = 1$, $U(a, s_2) = 0$, $U(b, s_2) = 3$. For $I = \emptyset$: (i) Preservation: $\mathbb{E}[U(a, S)] = 1$, $\mathbb{E}[U(b, S)] = 2$, so $\text{Opt}_{\emptyset}^{\text{stoch}} = \{b\}$; but $\text{Opt}(s_1) = \{a\}$,

so preservation fails. (ii) Decisiveness: The empty fiber itself has a unique conditional optimum $\{b\}$, so $I = \emptyset$ is decisive. This shows decisiveness is strictly stronger than preservation.

C. Stochastic Status at a Glance

Predicate/ query	What is proved here	Mechanized core	Open / not formalized
Stochastic sufficiency (preservation)	Polynomial-time under explicit-state encoding; bridges to static sufficiency	Explicit-state decider; full-support bridge package	No succinct-encoding hardness claimed
Minimum / anchor preservation	Polynomial-time in explicit state; under full support, inherit coNP-completeness / Σ_2^P -completeness from the static regime	Full-support inheritance theorems; finite searches; obstruction/bridge lemmas	Open; conjecture coNP-hard
Stochastic decisiveness	Polynomial-time under explicit-state encoding; PP-hard under succinct encoding	Reduction core; finite witness/checking schemata	No oracle-machine formalization
Minimum / anchor decisiveness	PP-hard; in NP^{PP} at paper level	Existential witness/checking schemata; bounded explicit-state searches	No oracle-class membership formalization

TABLE II: Regime classification summary for the stochastic row. Open entries mark explicit scope boundaries. Under full support, stochastic preservation is completely classified in the full-support base case.

The rest of this section unpacks the four rows of this summary. The preservation side is complete in explicit state and under full-support inheritance, and now also has support-sensitive partial results beyond full support. The decisiveness side carries the strongest succinct-encoding classification currently established in the paper.

Problem IV.6 (STOCHASTIC-SUFFICIENCY-CHECK). **Input:** $\mathcal{D}_S = (A, X_1, \dots, X_n, U, P)$ and $I \subseteq \{1, \dots, n\}$.

Question: Is I stochastically sufficient?

D. Stochastic Sufficiency

Theorem IV.7¹⁵ (Stochastic Sufficiency is Decidable in Explicit State). *Under the explicit-state encoding, STOCHASTIC-SUFFICIENCY-CHECK is decidable in polynomial time.*

¹⁵ : DC91–93, OU12

Proof. For a fixed information set I , the verifier scans the finite state space and checks whether the conditional fiber optimizer $\text{Opt}_I^{\text{stoch}}(s_I)$ agrees with the fully conditioned optimizer $\text{Opt}(s)$ at every state s . The artifact contains a counted explicit-state search witnessing exactly this predicate together with a linear-in- $|S|$ step bound¹⁶. Hence stochastic sufficiency is polynomial-time decidable in the explicit-state model. ■

a) Conjecture 4.1 (Support-Sensitive Preservation Complexity).: Under the succinct encoding of Section II-D, STOCHASTIC-PRESERVATION-CHECK for general distributions (without the full-support assumption) is coNP-hard.

Intuition: preservation remains syntactically universal (statewise optimizer agreement), so its quantifier profile aligns with coNP-style counterexample exclusion rather than PP-style majority comparison. The support-sensitive obstruction lemmas (Proposition IV.16) isolate why full-support transfer fails: zero-probability states can witness violations even when they never appear in conditional averages. The mechanized finite core for this obstruction is available¹⁷.

Open Problem 4.2. Determine the exact succinct-encoding complexity of STOCHASTIC-MINIMUM-PRESERVATION and STOCHASTIC-ANCHOR-PRESERVATION for general distributions. Under Conjecture IV-D0a, these inherit coNP-completeness and Σ_2^P -completeness respectively, matching the static regime; resolving the conjecture would complete the final open cell in the regime matrix (Table 2).

Proposition IV.8¹⁸ (Stochastic Sufficiency Implies Static Sufficiency). *If I is stochastically sufficient, then I is sufficient for the underlying static decision problem.*

Proof. If the coarse conditional optimizer agrees with the fully conditioned optimizer at every state, then any two states that agree on I also have equal full-information optimal-action sets. So the original static optimizer already factors through the projection to I . ■

Proposition IV.9¹⁹ (Full-Support Equivalence). *Assume the distribution P has full support and the action set is nonempty. Then I is statically sufficient if and only if it is stochastically sufficient.*

Proof. The forward direction uses full support to show that when the static optimizer is constant on each I -fiber, conditional averaging cannot change the optimizer on that fiber. The reverse direction is Proposition IV.8. Both directions are mechanized. ■

¹⁶ : DC91–93.

¹⁷ : OU12, DC96, DC98

¹⁸ : DC94

¹⁹ : DC95–96

Proposition IV.10²⁰ (Quotient Equivalence under Full Support). *Under the hypotheses of Proposition IV.9, the stochastic fiber quotient induced by I coincides with the original decision quotient.*

Proof. Under stochastic sufficiency, the conditional fiber optimizer and the fully conditioned optimizer agree statewise. Hence the induced stochastic equivalence relation matches the original decision-equivalence relation, so the quotient setoids coincide. Under full support, Proposition IV.9 supplies the needed preservation hypothesis. ■

Problem IV.11 (STOCHASTIC-MINIMUM-PRESERVATION). **Input:** $\mathcal{D}_S = (A, X_1, \dots, X_n, U, P)$ and $k \in \mathbb{N}$.

Question: Does there exist a stochastically sufficient coordinate set I with $|I| \leq k$?

Problem IV.12 (STOCHASTIC-ANCHOR-PRESERVATION). **Input:** $\mathcal{D}_S = (A, X_1, \dots, X_n, U, P)$ and $I \subseteq \{1, \dots, n\}$.

Question: Does there exist an anchor state s_0 such that for every state s agreeing with s_0 on I , one has $\text{Opt}_I^{\text{stoch}}(s_I) = \text{Opt}(s)$?

Theorem IV.13²¹ (Full-Support Inheritance for Preservation Variants). *Assume the distribution P has full support and the action set is nonempty. Then:*

- 1) STOCHASTIC-MINIMUM-PRESERVATION is equivalent to the underlying static MINIMUM-SUFFICIENT-SET query.
- 2) For each fixed I , STOCHASTIC-ANCHOR-PRESERVATION is equivalent to the underlying static ANCHOR-SUFFICIENCY query on I .

Consequently, under full support, stochastic minimum preservation is coNP-complete and stochastic anchor preservation is Σ_2^P -complete.

Proof. For minimum preservation, Proposition IV.9 applies pointwise to each candidate coordinate set I , so the existential minimization problem is unchanged under full support. For anchor preservation, if one anchor fiber preserves the full-information optimizer, then the optimizer is already constant on that fiber, giving static anchor sufficiency. Conversely, if the static optimizer is constant on one I -fiber, then under full support the same conditional-expectation argument used in Proposition IV.9 shows that the coarse conditional optimizer agrees with the full-information optimizer throughout that fiber. The complexity claims therefore inherit directly from Theorems III.5 and III.6. ■

²⁰ : DC97–99 ²¹ : DQ94–95

Proposition IV.14²² (Explicit Finite Search for Preservation Variants). *For finite instances, the artifact contains counted exhaustive-search procedures for stochastic minimum preservation and stochastic anchor preservation. Their certified step bounds are $O(2^n)$ and $O(|S|)$, respectively.*

Proof. The minimum-preservation search scans the subset lattice and invokes the explicit-state preservation checker on each candidate set. The anchor-preservation search scans candidate anchor states and checks preservation on the induced fiber. Both procedures are packaged with correctness theorems and explicit step bounds in the artifact. ■

Theorem IV.15²³ (Explicit-State Tractability for Preservation Variants). *Under the explicit-state encoding, STOCHASTIC-MINIMUM-PRESERVATION and STOCHASTIC-ANCHOR-PRESERVATION are both decidable in polynomial time.*

Proof. For minimum preservation, the explicit-state search enumerates coordinate subsets and invokes the explicit preservation checker on each candidate set; the resulting counted procedure is polynomial in the explicit input size for fixed state and coordinate budgets. For anchor preservation, the explicit-state search enumerates anchor states and checks preservation on the induced fiber, yielding a polynomial-time verifier. Both wrappers are mechanized in the artifact. ■

Proposition IV.16²⁴ (General-Distribution Obstructions and Support-Sensitive Bridge). *Let I be a coordinate set.*

- 1) *If I is stochastically sufficient, then I contains every relevant coordinate of the underlying static decision problem.*
- 2) *Consequently, in the Boolean-cube setting, if STOCHASTIC-MINIMUM-PRESERVATION has a yes-instance with budget k , then the static relevant-coordinate set has cardinality at most k .*
- 3) *If the underlying static decision problem is sufficient on I and every I -fiber contains at least one positive-probability state, then I is stochastically sufficient.*
- 4) *If the underlying static decision problem is anchor-sufficient on I and the anchor fiber contains a positive-probability state, then I is stochastically anchor-preserving.*

Proof. The first statement composes the unconditional bridge from stochastic preservation to static sufficiency with the static theorem that every sufficient set contains all relevant coordinates. The second is the resulting cardinality bound for any witness set in the minimum query. For the positive-support

²² : DQ86–89 ²³ : DQ92–93 ²⁴ : DQ90–91, DQ96–97

statements, the mechanized proof refines the full-support argument: it uses positivity only on one representative state per relevant fiber, together with fiberwise invariance of the conditional optimizer, to propagate static sufficiency or anchor sufficiency into stochastic preservation on the whole fiber. ■

b) What this paper establishes for preservation.: This yields a complete preservation classification under full support, together with support-sensitive partial theory beyond that base case. The PP-hardness results in the stochastic row concern decisiveness, not preservation.

Problem IV.17 (STOCHASTIC-DECISIVENESS-CHECK). **Input:** $\mathcal{D}_S = (A, X_1, \dots, X_n, U, P)$ and $I \subseteq \{1, \dots, n\}$.

Question: Is I stochastically decisive?

c) Naming convention.: From this point onward, the historical names “stochastic anchor sufficiency” and “stochastic minimum sufficiency” refer to the decisiveness family, not the preservation family introduced above.

Definition IV.18 (Stochastic Anchor Sufficiency). A coordinate set I is *stochastically anchor-sufficient* if there exists an admissible fiber value $\alpha \in \text{supp}(S_I)$ and an action $a \in A$ such that

$$\arg \max_{a' \in A} \mathbb{E}[U(a', S) \mid S_I = \alpha] = \{a\}.$$

Equivalently, some I -fiber has a unique conditional optimal action.

Problem IV.19 (STOCHASTIC-ANCHOR-SUFFICIENCY-CHECK). **Input:** $\mathcal{D}_S = (A, X_1, \dots, X_n, U, P)$ and $I \subseteq \{1, \dots, n\}$.

Question: Is I stochastically anchor-sufficient?

Problem IV.20 (STOCHASTIC-MINIMUM-SUFFICIENT-SET). **Input:** $\mathcal{D}_S = (A, X_1, \dots, X_n, U, P)$ and $k \in \mathbb{N}$.

Question: Does there exist a stochastically decisive coordinate set I with $|I| \leq k$?

E. Encoding-Sensitive Complexity of Stochastic Decisiveness

The stochastic regime preserves the same underlying certification template, namely whether the retained coordinates determine the optimizer, but asks it after conditioning and expectation have been introduced. The current complexity package concerns stochastic decisiveness because it admits a clean mechanized reduction core and directly supports the existential anchor/minimum queries. For the empty-information

case, one is already comparing aggregate mass of satisfying versus nonsatisfying states. More generally, stochastic decisiveness asks whether one action strictly dominates all competitors in conditional expected utility on each admissible fiber.

Theorem IV.21²⁵ (Stochastic Decisiveness is Encoding-Sensitive). *For the stochastic decisiveness predicate above, STOCHASTIC-DECISIVENESS-CHECK is decidable in polynomial time under the explicit-state encoding. Under the succinct encoding it is PP-hard. The accompanying upper-bound argument is proved in the paper via an explicit bad-fiber witness characterization whose finite witness/checking core is mechanized in the artifact.*

Proof. For the explicit-state encoding, the number of admissible fibers is at most $|S|$, so one can enumerate the fibers and, for each fiber value α , compute the conditional expected utilities of all actions directly from the table. Checking whether exactly one action attains the maximum on each fiber is therefore polynomial in the explicit input size.

For succinct-encoding hardness, reduce MAJSAT by the three-action gadget used in the mechanized reduction. Given Boolean formula φ , take $S = \{0, 1\}^n$, uniform P , actions $\{\text{accept}, \text{hold}_L, \text{hold}_R\}$, and utilities

$$U(\text{accept}, s) = \mathbf{1}[\varphi(s)], \quad U(\text{hold}_L, s) = U(\text{hold}_R, s) = \frac{1}{2} - 2^{-(n+1)},$$

with $I = \emptyset$. There is only one admissible fiber, so stochastic decisiveness asks whether the prior-optimal action is unique. By construction, this happens exactly when $\mathbb{E}[\varphi] \geq 1/2$, i.e., in the MAJSAT case. Thus MAJSAT many-one reduces to stochastic decisiveness.

For the upper bound, failure of decisiveness is witnessed by one bad state whose observed fiber does not have a singleton conditional optimum. The current artifact packages exactly this no-witness characterization together with the corresponding existential witness schema, and the two directions are bundled in a summary theorem²⁶ These mechanized witness/checking packages verify the finite combinatorial core used by the paper's oracle-class arguments. Concretely, the artifact verifies the existential-universal quantifier structure of the predicate, the witness-checking schema that guesses an anchor and verifies conditional-uniqueness using PP comparisons, and the finite-step-counted search wrappers that bound the nondeterministic guessing phase.²⁷ The standard oracle-machine reduction proving NP^{PP} membership

²⁵ : DC45, DC62, DC72, OU8-11 ²⁶ : OU8-11.

²⁷ : OU1-2, OU6-7

is argued in the paper text and not mechanized. The corresponding coNP^{PP} -style and NP^{PP} -style memberships are then proved in the paper text by standard complexity-theoretic reasoning. ■

Proposition IV.22²⁸ (Stochastic Decisiveness Yields Stochastic Anchor Sufficiency). *If I is stochastically decisive, then I is stochastically anchor-sufficient.*

Proof. Choose any admissible fiber value α . By stochastic decisiveness, there exists an action a such that

$$\text{Opt}_I^{\text{stoch}}(\alpha) = \{a\}.$$

That fiber therefore witnesses stochastic anchor sufficiency. ■

Theorem IV.23²⁹ (Stochastic Anchor Sufficiency is PP-hard). *STOCHASTIC-ANCHOR-SUFFICIENCY-CHECK is PP-hard.*

Proof. Reduce MAJSAT. Given Boolean formula φ on $n \geq 1$ variables, take $S = \{0, 1\}^n$, uniform P , $I = \emptyset$, and three actions $\{\text{accept}, \text{hold}_L, \text{hold}_R\}$. Set

$$U(\text{accept}, s) = \mathbf{1}[\varphi(s)], \quad U(\text{hold}_L, s) = U(\text{hold}_R, s) = \frac{1}{2} - 2^{-(n+1)}.$$

If at least half of all assignments satisfy φ , then accept is the unique optimal action on the empty-information fiber, so the instance is anchor-sufficient. If fewer than half satisfy φ , then hold_L and hold_R tie above accept , so no singleton anchor optimum exists. Thus MAJSAT many-one reduces to the stochastic anchor query. ■

Theorem IV.24³⁰ (Stochastic Minimum-Sufficient-Set is PP-hard). *STOCHASTIC-MINIMUM-SUFFICIENT-SET is PP-hard.*

Proof. Reduce MAJSAT to the $k = 0$ slice. In the same three-action gadget used above, there exists a stochastically decisive set of size at most 0 iff the empty set itself is stochastically decisive. This empty-set equivalence for the gadget is certified in the artifact, so MAJSAT many-one reduces to STOCHASTIC-MINIMUM-SUFFICIENT-SET. ■

Proposition IV.25³¹ (Explicit Finite Search for Stochastic Queries). *For finite instances, the artifact contains counted exhaustive-search procedures for stochastic sufficiency, stochastic decisiveness, stochastic*

²⁸ : DC19 ²⁹ : DC40–42 ³⁰ : DC45, DC47 ³¹ : DC62–65, DC72–73, DC76, DC91–93

anchor sufficiency, and stochastic minimum sufficiency. Their certified step bounds are respectively $O(|S|)$, $O(|S||A|)$, and $O(2^n)$.

Proof. The stochastic-sufficiency search scans for a state where the coarse conditional optimizer disagrees with the fully conditioned optimizer; the decisiveness search scans for a violating state whose fiber-optimal set is non-singleton; the anchor search scans over candidate anchor-state/action pairs; and the minimum query scans the subset lattice. Each procedure admits a counted boolean search formulation with both a correctness theorem and an explicit step bound. ■

Theorem IV.26³² (Stochastic Anchor and Minimum Queries Lie in NP^{PP}). *Under the present encoding, STOCHASTIC-ANCHOR-SUFFICIENCY-CHECK and STOCHASTIC-MINIMUM-SUFFICIENT-SET are both in NP^{PP} .*

Proof. For anchor sufficiency, the mechanized collapse

$$\begin{aligned} & \text{STOCHASTIC-ANCHOR-SUFFICIENCY-CHECK}(P, I) \\ & \iff \exists s_0 \exists a \left(\text{fiberOpt}(P, I, s_0) = \{a\} \right) \end{aligned}$$

is already available in the artifact.³³ A nondeterministic polynomial-time machine may therefore guess (s_0, a) and use a PP oracle to verify that a is the unique conditional optimum on the anchor fiber. Concretely, uniqueness can be checked by comparing the conditional expected utility of a against every competing action on that fiber, which is exactly the same majority-style conditional comparison used in the decisiveness verification.

For the minimum query, the machine guesses a coordinate set I with $|I| \leq k$ and then asks whether I is stochastically decisive. The same witness/checking template used for the anchor case yields the required PP-style verifier for the guessed set, so the minimum query is also in NP^{PP} . ■

Theorem IV.26 completes the complexity picture for stochastic decisiveness: decisiveness is polynomial-time in the explicit-state model and PP-hard in the succinct model, while the anchor and minimum decisiveness queries are PP-hard and lie in NP^{PP} . Those oracle-class memberships are proved in the paper text. The artifact independently verifies the finite core behind those proofs: the bad-fiber counter-witness schema for decisiveness, the anchor-collapse equivalence, the generic existential-witness schema,

³² : [OU1-2](#), [OU6-7](#), [PA3](#), [EH1-5](#) ³³ : [PA3](#)

bounded-witness recoveries for the explicit-state stochastic anchor and minimum searches, and existential-majority hardness/completeness packages for the anchor and decisiveness query families³⁴

a) Scope note.: For stochastic preservation, this paper establishes explicit-state tractability for the preservation family, full-support inheritance, quotient equivalence, and support-sensitive obstruction/bridge lemmas beyond full support. The general-distribution minimum and anchor variants still remain open as full classification problems and are not claimed. For stochastic decisiveness, the paper establishes PP-hardness and NP^{PP} membership under the stated encoding. The open entries in Table 2 are explicit scope boundaries rather than unresolved subcases.

The open entries in Table 2 are explicit scope boundaries. The gap between PP-hardness and NP^{PP} membership for anchor/minimum decisiveness is structural, not a bookkeeping omission; resolving it requires determining whether the outer existential layer can be absorbed into PP.

b) Relevance boundary.: The stochastic decisiveness query is fiber-indexed: fixing a candidate information set I induces a derived decision problem whose optimizer is the conditional fiber optimizer. In the current formalization this induced problem is always I -sufficient³⁵, but its relevance notion is therefore also indexed by I . For that reason, the paper does not assert a single global relevance-set characterization for stochastic minimum sufficiency analogous to the static and sequential regimes.

F. Tractable Subcases

Proposition IV.27³⁶ (Tractable Stochastic Cases). *Stochastic sufficiency is polynomial-time solvable under each of the following restrictions:*

- 1) *product distributions* $P = \bigotimes_i P_i$,
- 2) *bounded support* $|\text{supp}(P)| \leq k$,
- 3) *structured families where conditional expectations are computable in polynomial time (e.g. log-concave models with oracle access).*

Proof. In each case, conditional expected utilities can either be computed exactly or reduced to a polynomial number of marginal comparisons. ■

³⁴ : [OU8-11](#), [EH2-10](#).

³⁵ : [DC57](#)

³⁶ : [DQ13](#), [DQ15](#)

G. Bridge from Static

Distributional conditioning can genuinely destroy or preserve stochastic sufficiency depending on the structure of the input model.

Proposition IV.28³⁷ (Static Sufficiency Does Not Imply Stochastic Sufficiency). *There exist instances where I is sufficient in the static sense but not stochastically sufficient once a distribution P is fixed.*

Proof. Choose utilities where optimizer ties are harmless pointwise but are broken in expectation after conditioning. Then static fibers preserve Opt, while the conditional optimizer obtained from I fails to match the fully conditioned optimizer. ■

Proposition IV.29³⁸ (Static Singleton Sufficiency Transfers to Stochastic Sufficiency). *Suppose that on every I -fiber the static optimal-action set is a singleton, and that this singleton is constant on that fiber. Then I is stochastically sufficient for every distribution P .*

Proof. Fix an admissible fiber value α and let $\{a_\alpha\}$ be the common pointwise optimal-action set on that fiber. For every competing action $b \neq a_\alpha$ and every state s with $s_I = \alpha$, one has $U(a_\alpha, s) > U(b, s)$. Taking conditional expectations over the fiber preserves this strict inequality, so

$$\text{Opt}_I^{\text{stoch}}(\alpha) = \{a_\alpha\}.$$

Since this holds for every admissible fiber, the conditional optimizer on each fiber agrees with the fully conditioned optimizer at every state in that fiber. Hence I is stochastically sufficient. ■

The stochastic regime therefore gives the middle layer of the paper’s hierarchy: conditioning already raises exact certification beyond the static setting by turning exact certification into a counting comparison problem. Adding time will lift us further still: the next section shows how temporal contingency presents, alongside stochastic sufficiency, a decisiveness complexity family and then pushes the same underlying question from PP-style comparisons to PSPACE-level temporal certification.

V. SEQUENTIAL REGIME COMPLEXITY

We now move to sequential settings with transitions and observations. Here the same certification question is asked for a sequential decision model carrying transitions, observations, and a planning horizon. In the current formalization, the certified predicate is state-based: after packaging the sequential instance

³⁷ : DQ58 ³⁸ : DQ15–16

as its induced one-step decision map, can one suppress some coordinates without changing the resulting optimal-action set? This is where the complexity reaches PSPACE. The structural reason is that hidden information can matter later even when it appears irrelevant now. Certification must therefore account for temporally unfolding contingencies rather than compare only one-shot optimizers or conditional expectations. This complexity jump is consonant with the exact abstraction literature for temporally structured decision models: once hidden information can affect future contingent choices, exact aggregation and exact relevance certification both require reasoning about temporally mediated distinctions rather than one-shot optimizer comparisons. In succinct models, that temporal contingency supports the same kind of alternation that drives PSPACE hardness in planning and quantified-logic reductions.

A. Sequential Decision Problems

Definition V.1 (Sequential Decision Problem). A sequential decision problem is a tuple

$$\mathcal{D}_{\text{seq}} = (A, X_1, \dots, X_n, U, T, O),$$

with state space $S = X_1 \times \dots \times X_n$, transition kernel $T : A \times S \rightarrow \Delta(S)$, and observation model $O : S \rightarrow \Delta(\Omega)$.

Definition V.2 (Sequential Sufficiency). For a sequential instance $\mathcal{D}_{\text{seq}}$, let $\text{Opt}_{\text{seq}}(s)$ denote the optimal-action set of its induced decision map on state s . A coordinate set I is *sequentially sufficient* if

$$\forall s, s' \in S : \quad s_I = s'_I \implies \text{Opt}_{\text{seq}}(s) = \text{Opt}_{\text{seq}}(s').$$

This is the sequential state-based certification predicate used throughout the paper and in the artifact.

a) Two interpretations.: There are two closely related readings of this definition. Formally, the sequential row studies sufficiency for the optimizer induced by the sequential model. Its intended interpretation is policy-level: hidden information may be irrelevant at time zero and still become relevant later because it changes future contingent choices.

One can also read the same construction heuristically as a coordinate-restricted POMDP abstraction question. A full controller conditions on evolving observations and therefore on induced beliefs over latent states. Hiding coordinates changes the informational interface through which future decisions are made. That heuristic is useful for intuition and for understanding the TQBF gadget, but the formal theorems in

this paper are stated for the state-based predicate above rather than for a fully developed history-dependent policy class.

b) Scope and research direction.: The sequential model studied here is a state-indexed product model, not a POMDP with belief-state dynamics or an MDP with policy optimization. This captures the case where each time step’s decision is governed by a time-indexed utility, and the question is which cross-temporal coordinates matter. Extending the optimizer quotient to belief-state POMDPs and history-dependent policies, where the quotient would be over belief states rather than world states and the complexity classification would need to engage with the policy optimization layer, remains a natural research direction.

Problem V.3 (SEQUENTIAL-SUFFICIENCY-CHECK). **Input:** $\mathcal{D}_{\text{seq}}$ and $I \subseteq \{1, \dots, n\}$.

Question: Is I sequentially sufficient?

Definition V.4 (Sequential Anchor Sufficiency). A coordinate set I is *sequentially anchor-sufficient* if there exists an anchor state whose I -agreement class preserves the optimal action set throughout that class.

Problem V.5 (SEQUENTIAL-ANCHOR-SUFFICIENCY-CHECK). **Input:** $\mathcal{D}_{\text{seq}}$ and $I \subseteq \{1, \dots, n\}$.

Question: Is I sequentially anchor-sufficient?

Problem V.6 (Sequential Minimum Query). **Input:** $\mathcal{D}_{\text{seq}}$ and $k \in \mathbb{N}$.

Question: Does there exist a sequentially sufficient coordinate set I with $|I| \leq k$?

Proposition V.7³⁹ (Sequential Minimality Equals Relevance). *For any minimal sequentially sufficient set I , a coordinate lies in I if and only if it is relevant for the underlying decision boundary.*

Proof. In the current formalization, sequential sufficiency is defined as sufficiency of the underlying one-step optimizer map on the state space. The static minimal-set/relevance equivalence therefore transports directly to the sequential setting. ■

B. PSPACE-Completeness

The sequential regime is the strongest of the three because hidden information can matter not only for the current optimizer but for the future evolution of the decision problem. Static certification asks whether bad state pairs exist; stochastic certification asks whether conditional averages cross a threshold; sequential certification asks whether omitted coordinates can change the induced optimizer after temporal

³⁹ : DC49–50

structure is taken into account. That added temporal contingency is exactly what the TQBF reduction exploits.

The reduction follows the same structural pattern as quantified formulas and temporally structured control problems elsewhere in the planning literature. Existential quantifier blocks correspond to controller-side choices, universal blocks correspond to adversarial or environment-controlled resolutions, and the truth value of the quantified formula is encoded into the induced optimizer of the reduced sequential instance. A hidden coordinate can look locally irrelevant after one transition yet still encode which branch of the quantified game has been entered. The reduction exposes the source of PSPACE hardness: exact relevance certification must reason about information whose effect is mediated by temporal contingency rather than by a single one-shot comparison.

Theorem V.8⁴⁰ (Sequential Sufficiency is PSPACE-complete). *SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete.*

Proof. For membership, the verifier ranges over state pairs or candidate witnesses while evaluating the induced sequential optimizer of the input instance. Under the present finite-horizon encoding, that induced optimizer can be evaluated using only polynomial space in the succinct input description of the reduction gadgets, and the outer search can likewise be carried out in polynomial space. This is the verifier pattern abstracted by the explicit finite-search layer formalized in the artifact.

For hardness, reduce TQBF. Encode alternating quantifiers into the reduced sequential instance so that the induced optimizer is state-independent exactly when the quantified instance is true. Then empty-set sequential sufficiency holds exactly when the source TQBF instance is true. ■

The proof can be read operationally as follows. A sequential certificate must rule out the possibility that some hidden bit becomes decision-relevant only after several temporally mediated updates. That means the verifier cannot stop at a single static comparison; it has to account for contingent future structure in the induced optimizer. TQBF is therefore the correct reduction target: the same alternation between “there exists a choice” and “for every resolution” is already built into the reduced sequential instance.

Proposition V.9⁴¹ (Sequential Sufficiency Refines to Sequential Anchor Sufficiency). *If I is sequentially sufficient, then I is sequentially anchor-sufficient.*

Proof. If optimal-action sets are preserved for every pair of states agreeing on I , then fixing any anchor state immediately yields preservation across its entire I -agreement class. ■

⁴⁰ : DC46, DC66, DC74 ⁴¹ : DC23

Theorem V.10⁴² (Sequential Anchor Query is PSPACE-complete). *The sequential anchor query is PSPACE-complete.*

Proof. Membership is in PSPACE. By definition, the query asks whether there exists an anchor state s_0 such that every state agreeing with s_0 on I preserves the same optimal-action set. A nondeterministic polynomial-space procedure can guess s_0 and then scan candidate states one at a time, verifying the agreement condition and comparing the corresponding optimal-action sets. Since $\text{NPSPACE} = \text{PSPACE}$, the query lies in PSPACE.

Reuse the TQBF reduction for sequential sufficiency with $I = \emptyset$. At empty information, sequential anchor sufficiency is equivalent to global preservation of the optimal-action set, so the same construction yields

$$\text{TQBF}(q) \iff \text{SEQUENTIAL-ANCHOR-SUFFICIENCY-CHECK}(\text{reduceTQBF}(q), \emptyset).$$

Hence TQBF many-one reduces to the sequential anchor query. ■

Theorem V.11⁴³ (Sequential Minimum Query is PSPACE-complete). *The sequential minimum-sufficient-set query is PSPACE-complete.*

Proof. Membership is in PSPACE. The query asks whether there exists a coordinate set I of size at most k such that I is sequentially sufficient. A nondeterministic polynomial-space procedure can guess I and then run the same polynomial-space verifier used for sequential sufficiency. Again, $\text{NPSPACE} = \text{PSPACE}$.

Reduce TQBF to the $k = 0$ slice. There exists a sequentially sufficient set of size at most 0 iff the empty set itself is sequentially sufficient. The same construction yields PSPACE-hardness for SEQUENTIAL-MINIMUM-SUFFICIENT-SET. ■

Proposition V.12⁴⁴ (Explicit Finite Search for Sequential Queries). *For finite instances, the development contains counted exhaustive-search procedures for sequential sufficiency, sequential anchor sufficiency, and sequential minimum sufficiency. Their certified step bounds are respectively $O(|S|^2)$, $O(|S|)$, and $O(2^n)$.*

Proof. The sufficiency search scans for an agreeing pair of states with different optimal sets; the anchor search scans over candidate anchor states; and the minimum query scans the subset lattice. Each procedure

⁴² : DC28–29, DC44 ⁴³ : DC46, DC48 ⁴⁴ : DC66–69, DC74–76

admits a counted boolean search formulation with both a correctness theorem and an explicit step bound.

■

C. Tractable Subcases

Proposition V.13⁴⁵ (Tractable Sequential Cases). *The sequential sufficiency problem is polynomial-time solvable under each of the following restrictions:*

- 1) *full observability (MDP case),*
- 2) *bounded horizon,*
- 3) *tree-structured transition systems,*
- 4) *deterministic transitions.*

Proof. Each restriction reduces policy search to dynamic programming over a polynomially bounded representation. ■

D. Bridge from Stochastic

The bridge from the stochastic regime is strict. A one-shot distribution can hide no temporal contingency; once dynamics are introduced, latent information can become decision-relevant later even when it appears irrelevant at time zero.

Proposition V.14⁴⁶ (Stochastic Sufficiency Does Not Imply Sequential Sufficiency). *There exist instances where I is sufficient in the stochastic one-shot sense but fails to be sufficient once temporal dynamics are introduced.*

Proof. Construct a process in which the same one-step expected optimizer is preserved under I , but hidden-state memory affects future information states and therefore optimal policies. The temporal dependence blocks one-shot sufficiency transfer. ■

With the static, stochastic, and sequential classifications in place, the regime hierarchy is now complete. The next section consolidates that hierarchy into a single matrix, and the following section refines it with the encoding-sensitive ETH lower bounds that explain when exact certification remains feasible despite the worst-case class barriers.

⁴⁵ : DQ12, DQ14, DQ45 ⁴⁶ : DQ62

VI. REGIME MATRIX

The static, stochastic, and sequential formulations are the paper’s central organizing object. They ask the same underlying question about which coordinates matter for the decision, but they answer it in increasingly expressive models. In the stochastic regime, preservation and decisiveness become parallel exact predicates. Adding time then introduces temporal contingency. Read as a complexity matrix, adding probability introduces conditional-comparison hardness, and adding time lifts the core queries further to PSPACE.

A. Complexity Matrix

Regime / predicate	Base query	Minimum	Anchor
Static	coNP-complete	coNP-complete	Σ_2^P -complete
Stochastic preservation	P (explicit), bridge theorems	P (explicit); coNP-complete (full) / open (general)	P (explicit); Σ_2^P -complete (full) / open (general)
Stochastic decisiveness	P (explicit), PP-hard (succinct)	PP-hard, in NP^{PP}	PP-hard, in NP^{PP}
Sequential	PSPACE-complete	PSPACE-complete	PSPACE-complete

TABLE III: Complexity matrix across regimes and query types.

Table III is the main statement-level summary. The stochastic row is intentionally split: preservation and decisiveness are distinct predicates with different complexity behavior. Open cells are explicit scope boundaries.

Theorem VI.1⁴⁷ (Regime-Sensitive Complexity Matrix). *Under the encoding conventions of Section II-D, the classification is as follows. In the static regime, sufficiency and minimum sufficiency are coNP-complete and anchor sufficiency is Σ_2^P -complete. In the stochastic regime, preservation (stochastic sufficiency) has a polynomial-time base query under explicit-state encoding with proved bridges back to static sufficiency; the minimum and anchor preservation variants are also polynomial-time under explicit-state encoding, are coNP-complete and Σ_2^P -complete under full support, and satisfy the support-sensitive partial results of Proposition IV.16. Decisiveness (stochastic decisiveness) has a polynomial-time base query under explicit-state encoding and is PP-hard under the succinct encoding; the anchor and minimum decisiveness queries are PP-hard and lie in NP^{PP} , with those oracle-class memberships proved in the paper text and their finite*

⁴⁷ : DC100, DQ1, DQ17–19, OU1–2, OU11–12, EH4–10

witness/checking cores mechanized. In the sequential regime, sufficiency, minimum, and anchor queries are PSPACE-complete.

Proof. Immediate by table lookup from Theorems III.4, III.5, III.6, IV.7, IV.13, IV.15, Proposition IV.16, IV.21, IV.23, IV.24, IV.26, V.8, V.10, and V.11. ■

B. Strict Regime Gaps

Theorem VI.2⁴⁸ (Static-to-Stochastic Complexity Gap). *Assuming $\text{coNP} \neq \text{PP}$, the succinct stochastic decisiveness classification is strictly harder in worst-case complexity than the static sufficiency classification.*

Proof. Static sufficiency checking is coNP -complete (Theorem III.4), while stochastic decisiveness is PP -hard under the succinct encoding (Theorem IV.21). Under the assumption $\text{coNP} \neq \text{PP}$, this yields a strict complexity gap between the two classifications. ■

Theorem VI.3⁴⁹ (Stochastic-to-Sequential Complexity Gap). *Assuming $\text{PP} \neq \text{PSPACE}$, the sequential sufficiency classification is strictly harder in worst-case complexity than the stochastic decisiveness classification.*

Proof. Stochastic decisiveness is PP -hard under the succinct encoding (Theorem IV.21), while sequential sufficiency is PSPACE -complete (Theorem V.8). The class gap yields a strict complexity separation between the two classifications under the assumption $\text{PP} \neq \text{PSPACE}$. ■

a) Forward-looking perspective.: The regime hierarchy extends to richer models: partially observable settings, multi-agent decision problems, and game-theoretic regimes would introduce additional layers of complexity such as knowledge reasoning or equilibrium computation. The optimizer-quotient framework extends to those regimes, though their detailed classification remains open.

VII. ENCODING DICHOTOMY AND ETH LOWER BOUNDS

The regime hierarchy identifies where exact relevance certification sits in the standard complexity landscape. This section extracts the complementary algorithmic message: under the encodings of Section II-D, there is a strong contrast between explicit-state instances whose true relevant support is small and succinct instances whose relevant support is extensive. The former admit direct algorithms; the latter inherit exponential lower bounds under ETH. The practical takeaway is cross-model rather

⁴⁸ : DQ18 ⁴⁹ : DQ19

than single-model: logarithmic relevant support can still be certified exactly in polynomial time under explicit encodings, while linear relevant support under succinct encodings already forces ETH-conditioned exponential cost.

a) Model note.: Part 1 is an explicit-state upper bound (time polynomial in $|S|$). Part 2 is a succinct-encoding lower bound under ETH (time exponential in n). The encodings are defined in Section II-D.

Encoding	Support regime	Complexity consequence	Source
Explicit-state	$k^* = O(\log N)$	exact certification is polynomial in N by projection enumeration	Theorem VII.1(1)
Succinct	$k^* = \Omega(n)$	exact certification inherits an ETH-conditioned $2^{\Omega(n)}$ lower bound	Theorem VII.1(2)

TABLE IV: Boundary summary. The tractable-hard contrast is controlled jointly by encoding and by the size of the true sufficient support.

Theorem VII.1⁵⁰ (Encoding-Sensitive Contrast). *Let $\mathcal{D} = (A, X_1, \dots, X_n, U)$ be a decision problem with $|S| = N$ states. Let k^* be the size of the minimal sufficient set.*

- 1) **Logarithmic case (explicit-state upper bound):** *If $k^* = O(\log N)$, then SUFFICIENCY-CHECK is solvable in polynomial time in N under the explicit-state encoding.*
- 2) **Linear case (succinct lower bound under ETH):** *If $k^* = \Omega(n)$, then SUFFICIENCY-CHECK requires time $\Omega(2^{n/c})$ for some constant $c > 0$ under the succinct encoding (assuming ETH).*

Proof. Part 1 (Logarithmic case): If $k^* = O(\log N)$, then the number of distinct projections $|S_{I^*}|$ is at most $2^{k^*} = O(N^c)$ for some constant c . Under the explicit-state encoding, we enumerate all projections and verify sufficiency in polynomial time.

Part 2 (Linear case): We establish this by a standard ETH transfer under the succinct encoding. ■

A. The ETH Reduction Chain

The lower bound in Part 2 of Theorem VII.1 follows from a short standard ETH transfer. Under ETH, 3-SAT requires $2^{\Omega(n)}$ time. Negation gives a linear-time reduction from 3-SAT to TAUTOLOGY. The static reduction of Theorem III.4 then maps an n -variable tautology instance to a sufficiency instance with the same asymptotic coordinate count. Therefore, if SUFFICIENCY-CHECK were solvable in $2^{o(n)}$ time under the succinct encoding, then so would TAUTOLOGY and hence 3-SAT, contradicting ETH.

Definition VII.2 (Exponential Time Hypothesis (ETH)). There exists a constant $\delta > 0$ such that 3-SAT on n variables cannot be solved in time $O(2^{\delta n})$ [1].

⁵⁰ : DQ26, DQ29–30

Proposition VII.3⁵¹ (Tight Constant). *The reduction in Theorem III.4 preserves the number of variables up to an additive constant: an n -variable formula yields an $(n + 1)$ -coordinate decision problem. Therefore, the constant c in the $2^{n/c}$ lower bound is asymptotically 1:*

$$\text{SUFFICIENCY-CHECK requires time } \Omega(2^{\delta(n-1)}) = 2^{\Omega(n)} \text{ under ETH}$$

where δ is the ETH constant for 3-SAT.

Proof. The TAUTOLOGY reduction (Theorem III.4) constructs:

- State space $S = \{\text{ref}\} \cup \{0, 1\}^n$ with $n + 1$ coordinates (one extra for the reference state)
- Query set $I = \emptyset$

When φ has n variables, the constructed problem has $n + 1$ coordinates. The asymptotic lower bound is $2^{\Omega(n)}$ with the same constant δ from ETH. ■

B. Cross-Model Contrast

Corollary VII.4⁵² (Cross-Model Contrast for Exact Certification). *Across the explicit-state and succinct encodings of Section II-D, exact certification exhibits the following contrast:*

- If $k^* = O(\log N)$, SUFFICIENCY-CHECK is polynomial in N under the explicit-state encoding
- If $k^* = \Omega(n)$, SUFFICIENCY-CHECK is exponential in n under ETH in the succinct encoding

For Boolean coordinate spaces ($N = 2^n$), one has $\log N = n$. So the explicit-state upper bound should be read as polynomial in the explicit table size when $k^* = O(n)$, while the succinct lower bound is exponential in the succinct variable count when $k^* = \Omega(n)$. The comparison is therefore between two encodings of the same Boolean family, not between $O(\log n)$ and $\Omega(n)$ inside a single model.

Proof. The logarithmic case (Part 1 of Theorem VII.1) gives polynomial time when $k^* = O(\log N)$. More precisely, when $k^* \leq c \log N$ for constant c , the algorithm runs in time $O(N^c \cdot \text{poly}(n))$.

The linear case (Part 2) gives exponential time when $k^* = \Omega(n)$.

The transition point inside the explicit-state bound is where $2^{k^*} = N^{k^*/\log N}$ stops being polynomial in N , i.e., when $k^* = \omega(\log N)$. For Boolean coordinate spaces, however, the present theorem should be read only as a cross-encoding contrast, because $\log N = n$ and the succinct lower bound is parameterized directly by the succinct variable count. ■

⁵¹ : DQ27, DQ30 ⁵² : DQ3, DQ26

Remark VII.5 (Asymptotic Tightness Within Each Encoding). Under ETH, the lower bound is asymptotically tight in the succinct encoding. The explicit-state upper bound is tight in the sense that it matches enumeration complexity in N .

This encoding-sensitive contrast explains why exact simplification can be tractable on some instance families and computationally prohibitive on others. When the true sufficient support is tiny and the state space is explicit, exhaustive certification can still be polynomial. When the true support is extensive and the problem is given succinctly, the same certification task inherits a worst-case $2^{\Omega(n)}$ lower bound under ETH in addition to the class-level coNP hardness statement.

a) Interpretation.: The ETH chain composes with the structural rank viewpoint introduced earlier. When relevance is extensive, the same objects that drive the regime hierarchy also force the exponential obstruction in the succinct model. In that sense, the dichotomy theorem complements the hierarchy rather than standing apart from it: the hierarchy locates exact certification in the standard class landscape, while the dichotomy explains when explicit structure can still make exact certification feasible.

Remark VII.6 (Circuit Model Formalization). The ETH lower bound is stated in the succinct circuit encoding of Section II-D, where the utility function $U : A \times S \rightarrow \mathbb{R}$ is represented by a Boolean circuit computing $\mathbf{1}[U(a, s) > \theta]$ for threshold comparisons. In this model:

- The input size is the circuit size m , not the state space size $|S| = 2^n$
- A 3-SAT formula with n variables and c clauses yields a circuit of size $O(n + c)$
- The reduction preserves instance size up to constant factors: $m_{\text{out}} \leq 3 \cdot m_{\text{in}}$

This linear size preservation is essential for ETH transfer. In the explicit enumeration model (where S is given as a list), the reduction would blow up the instance size exponentially, precluding ETH-based lower bounds. The circuit model is standard in fine-grained complexity and matches practical representations of decision problems.

VIII. TRACTABLE SPECIAL CASES

We distinguish the encodings of Section II-D. The tractability results below state the model assumption explicitly.

Our theory provides comprehensive coverage of tractable subcases. The artifact certifies explicit decision procedures for several families, and also certifies paper-specific reductions and complexity transfers connecting structurally interesting restrictions to standard polynomial-time algorithmic backbones. Each

subcase removes one of the structural sources of hardness in exact certification: unrestricted action comparison, cross-coordinate interaction, high-width dependency structure, or unnecessary state-space multiplicity.

Restriction	Algorithmic mechanism	Complexity
Bounded actions	brute-force enumeration is polynomial in $ S $ when $ A $ constant	$O(S ^2 \cdot k^2)$
Separable utility	optimal action independent of state	$O(1)$
Low tensor rank	factored evaluation avoids full state enumeration	$O(A \cdot R \cdot n)$
Tree structure	dynamic programming over dependency tree suffices	$O(n \cdot A \cdot k_{\max})$
Bounded treewidth	interaction checking reduces to low-width CSP	$O(n \cdot k^{w+1})$
Coordinate symmetry	orbit types replace raw state pairs	$O\left(\binom{d+k-1}{k-1}^2\right)$

TABLE V: Tractability map. Each restriction above removes a structural hardness source. Formal definitions for all cases are retained in the subsections below.

Theorem VIII.1 (Tractable Subcases). *The following structurally interesting tractable subcases are all mechanized in artifact. The artifact certifies paper-specific reductions and complexity transfers connecting each to standard polynomial-time algorithmic backbones. Formal definitions and trivial cases (single action, bounded state space, separable utility, dominance) are retained in the subsections below.*

- 1) **Low tensor rank:** *The utility admits a rank- R decomposition $U(a, s) = \sum_{r=1}^R w_r \cdot f_r(a) \cdot \prod_i g_{ri}(s_i)$. The artifact certifies reduction to factored computation with bound $O(|A| \cdot R \cdot n)$.⁵³*
- 2) **Tree-structured dependencies:** *Coordinates are ordered such that s_i depends only on $(s_1, \dots, s_{i-1})$. The artifact certifies an explicit sufficiency checker for this class.⁵⁴*
- 3) **Bounded treewidth:** *The interaction graph has treewidth $\leq w$ and utility factors over edges. The artifact certifies reduction to bounded-treewidth CSP together with bound $O(n \cdot k^{w+1})$ inherited from standard CSP algorithm.⁵⁵*
- 4) **Coordinate symmetry:** *Utility is invariant under coordinate permutations. The artifact certifies reduction to orbit-type representatives together with resulting orbit-count bound $O\left(\binom{d+k-1}{k-1}^2\right)$.⁵⁶*

The following subsections provide the formal definitions and reduction theorems for each tractable subcase.

⁵³ : DQ77.

⁵⁴ : DQ63.

⁵⁵ : DQ81.

⁵⁶ : DQ85.

A. Bounded Actions

When the action set is bounded, brute-force enumeration becomes polynomial.

Definition VIII.2 (Bounded Action Problem). A decision problem has *bounded actions* with parameter k if $|A| \leq k$ for some constant k independent of the state space size.

Theorem VIII.3⁵⁷ (Bounded Actions Tractability). *If $|A| \leq k$ for constant k , then SUFFICIENCY-CHECK runs in $O(|S|^2 \cdot k^2)$ time.*

Proof. Given the full table of $U(a, s)$:

- 1) Compute $\text{Opt}(s)$ for all $s \in S$ in $O(|S| \cdot k)$ time.
- 2) For each pair (s, s') with $s_I = s'_I$, compare $\text{Opt}(s)$ and $\text{Opt}(s')$ in $O(k^2)$ time.
- 3) Total: $O(|S|^2)$ pairs $\times O(k^2)$ comparisons = $O(|S|^2 \cdot k^2)$.

When k is constant, this is polynomial in $|S|$. ■

B. Separable Utility (Rank 1)

Separable utility decouples actions from states, making sufficiency trivial.

Definition VIII.4 (Separable Utility). A utility function is *separable* if there exist functions $f : A \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ such that $U(a, s) = f(a) + g(s)$ for all $(a, s) \in A \times S$.

Theorem VIII.5⁵⁸ (Separable Tractability). *If utility is separable, then $I = \emptyset$ is always sufficient.*

Proof. If $U(a, s) = f(a) + g(s)$, then:

$$\text{Opt}(s) = \arg \max_{a \in A} [f(a) + g(s)] = \arg \max_{a \in A} f(a)$$

The optimal action depends only on f , not on s . Hence $\text{Opt}(s) = \text{Opt}(s')$ for all s, s' , making any I (including $\emptyset$) sufficient. ■

C. Low Tensor Rank

Utility functions with low tensor rank admit efficient factored computation, generalizing separable utility.

Definition VIII.6 (Tensor Rank Decomposition). A utility function $U : A \times ((i : \text{Fin } n) \rightarrow X_i) \rightarrow \mathbb{R}$ has *tensor rank* R if there exist weights $w_1, \dots, w_R \in \mathbb{R}$, action factors $f_r : A \rightarrow \mathbb{R}$, and coordinate factors $g_{ri} : X_i \rightarrow \mathbb{R}$ such that:

$$U(a, s) = \sum_{r=1}^R w_r \cdot f_r(a) \cdot \prod_{i=1}^n g_{ri}(s_i)$$

⁵⁷ : [DQ51](#), [DQ72](#) ⁵⁸ : [DQ57](#), [DQ74](#)

Remark VIII.7. Separable utility (Definition VIII.4) is the special case $R = 1$ with $w_1 = 1$, $f_1(a) = f(a)$, and $\prod_i g_{1i}(s_i) = g(s)$.

Theorem VIII.8⁵⁹ (Low Tensor Rank Reduction). *If utility has tensor rank R , the artifact certifies a reduction of the relevant optimization step to factored computation with bound $O(|A| \cdot R \cdot n)$.*

Proof. The artifact certifies three ingredients. First, bounded-rank tensor contraction admits a polynomial bound⁶⁰ Second, the low-rank utility representation reduces the relevant optimization step to factored computation in $O(|A| \cdot R \cdot n)$ steps⁶¹ Third, these are exactly the paper-specific ingredients needed to invoke the standard tensor-network algorithms cited in the text. Thus the formal support here is a certified reduction-and-transfer statement rather than an end-to-end formalization of the tensor-network backbone itself. ■

D. Tree-Structured Dependencies

When coordinate dependencies form a tree, dynamic programming yields polynomial-time sufficiency checking.

Definition VIII.9 (Tree-Structured Dependencies). A decision problem has *tree-structured dependencies* if there exists an ordering of coordinates $(1, \dots, n)$ such that the utility decomposes as:

$$U(a, s) = \sum_{i=1}^n u_i(a, s_i, s_{\text{parent}(i)})$$

where $\text{parent}(i) < i$ for all $i > 1$, and the root term depends only on (a, s_1) .

Theorem VIII.10⁶² (Tree Structure Tractability). *If dependencies are tree-structured with explicit local tables, SUFFICIENCY-CHECK runs in $O(n \cdot |A| \cdot k_{\max})$ time where $k_{\max} = \max_i |X_i|$.*

Proof. Dynamic programming on the tree order:

- 1) Process coordinates in order $i = n, n-1, \dots, 1$.
- 2) For each node i and each value of its parent coordinate, compute the set of actions optimal for some subtree assignment.
- 3) Combine child summaries with local tables in $O(|A| \cdot k_{\max})$ per node.
- 4) Total: $O(n \cdot |A| \cdot k_{\max})$.

A coordinate is relevant iff varying its value changes the optimal action set. ■

⁵⁹ : DQ75–77 ⁶⁰ : DQ75.

⁶¹ : DQ76–77.

⁶² : DQ63, DQ78

E. Bounded Treewidth

The tree-structured case generalizes to bounded treewidth interaction graphs via CSP reduction. This directly parallels backdoor tractability methods for CSPs [2], where finding a small backdoor to a tractable class enables efficient solving.

Definition VIII.11 (Interaction Graph). Given a pairwise utility decomposition, the *interaction graph* $G = (V, E)$ has:

- Vertices $V = \{1, \dots, n\}$ (one per coordinate)
- Edges $E = \{(i, j) : i \text{ and } j \text{ interact in utility}\}$

Definition VIII.12 (Pairwise Utility). A utility function is *pairwise* if it decomposes as:

$$U(a, s) = \sum_{i=1}^n u_i(a, s_i) + \sum_{(i,j) \in E} u_{ij}(a, s_i, s_j)$$

with unary terms u_i and binary terms u_{ij} given explicitly.

Theorem VIII.13⁶³ (Bounded Treewidth Reduction). *If the interaction graph has treewidth $\leq w$ and utility is pairwise with explicit factors, the artifact certifies a reduction to bounded-treewidth CSP together with standard complexity bound $O(n \cdot k^{w+1})$, where $k = \max_i |X_i|$. The reduction bridges to established CSP algorithms [2]–[4].*

Proof. The artifact certifies the paper-specific reduction from pairwise sufficiency checking to a CSP on the interaction graph⁶⁴ It also certifies the transfer of the standard bounded-treewidth complexity bound⁶⁵ Thus formal support here is a certified reduction-and-transfer theorem: the bounded-treewidth algorithm itself is standard (drawing on backdoor tractability methods [2]), while the paper-specific bridge to that algorithm is mechanized. ■

F. Coordinate Symmetry

When utility is invariant under coordinate permutations, the effective state space collapses to orbit types.

Definition VIII.14 (Dimensional State Space). A *dimensional state space* with alphabet size k and dimension d is $S = \{0, \dots, k-1\}^d$, the set of d -tuples over a k -element alphabet.

⁶³ : DQ79–81 ⁶⁴ : DQ80.

⁶⁵ : DQ79, DQ81.

Definition VIII.15 (Symmetric Utility). A utility function on a dimensional state space is *symmetric* if it is invariant under coordinate permutations: for all permutations $\sigma \in \mathfrak{S}_d$ and states $s \in S$,

$$U(a, s) = U(a, \sigma \cdot s) \quad \text{where } (\sigma \cdot s)_i = s_{\sigma^{-1}(i)}$$

Definition VIII.16 (Orbit Type). The *orbit type* of a state $s \in \{0, \dots, k-1\}^d$ is the multiset of its coordinate values:

$$\text{orbitType}(s) = \{\{s_1, s_2, \dots, s_d\}\}$$

Two states have the same orbit type iff they lie in the same orbit under coordinate permutation.

Theorem VIII.17⁶⁶ (Orbit Type Characterization). *For dimensional state spaces, two states s, s' have equal orbit types if and only if there exists a coordinate permutation σ such that $\sigma \cdot s = s'$.*

Proof. If: Permuting coordinates preserves the multiset of values. **Only if:** If s and s' have the same multiset of values, we can construct a permutation mapping each occurrence in s to the corresponding occurrence in s' . ■

Theorem VIII.18⁶⁷ (Symmetric Sufficiency Reduction). *Under symmetric utility, optimal actions depend only on orbit type:*

$$\text{orbitType}(s) = \text{orbitType}(s') \implies \text{Opt}(s) = \text{Opt}(s')$$

Thus SUFFICIENCY-CHECK reduces to checking pairs with different orbit types.

Proof. If $\text{orbitType}(s) = \text{orbitType}(s')$, then by Theorem VIII.17 there exists σ with $\sigma \cdot s = s'$. By symmetry, $U(a, s) = U(a, s')$ for all a , so $\text{Opt}(s) = \text{Opt}(s')$.

For sufficiency, we need only check pairs (s, s') agreeing on I with *different* orbit types; same-orbit pairs are automatically equal. ■

Theorem VIII.19⁶⁸ (Symmetric Complexity Bound). *Under symmetric utility, the artifact certifies that the number of orbit types is bounded by the stars-and-bars count:*

$$|\text{OrbitTypes}| = \binom{d+k-1}{k-1}$$

and that sufficiency checking reduces to cross-orbit comparisons. Thus the effective number of representative pair checks is at most $\binom{d+k-1}{k-1}^2$, which is polynomial in d for fixed k .

⁶⁶ : DQ82 ⁶⁷ : DQ83–84 ⁶⁸ : DQ56, DQ85

Proof. An orbit type is a multiset of d values from $\{0, \dots, k-1\}$, equivalently a k -tuple of non-negative integers summing to d . By stars-and-bars, the count is $\binom{d+k-1}{k-1}$. The mechanized content here is the orbit-type reduction and the resulting representative-count bound; the resulting polynomial-time conclusion then follows from this certified compression of the comparison space.

For fixed k , this is $O(d^{k-1})$, polynomial in d . ■

G. Practical Implications

The formally developed tractable subcases correspond to common modeling scenarios. The structurally nontrivial cases each remove a distinct source of hardness:

- **Bounded actions:** Small action sets (e.g., binary decisions, few alternatives). Brute-force enumeration is polynomial when $|A|$ is constant.
- **Separable utility:** Additive cost models, linear utility functions. The optimizer is independent of state, making any coordinate set sufficient.
- **Low tensor rank:** Factored preference models, low-rank approximations in recommendation systems or multi-attribute utility theory.
- **Tree structure:** Hierarchical decision processes, causal models with chain or tree dependency. Dynamic programming replaces exhaustive search.
- **Bounded treewidth:** Spatial models, graphical models with limited interaction width. CSP algorithms on low-treewidth graphs handle the sufficiency check.
- **Coordinate symmetry:** Exchangeable features, order-invariant utilities (e.g., set-valued inputs). Orbit-type representatives collapse the comparison space.

Simple edge cases also yield immediate tractability without structural argument: single action ($|\text{Opt}| = |A|$ trivially), bounded state space (exhaustive search is polynomial in $|S|$), strict dominance (one action always weakly optimal), and constant optimal set ($\text{Opt}(s)$ independent of s).

For problems given in the succinct encoding without these structural restrictions, the hardness results of Section III apply, justifying heuristic approaches.

IX. THE CERTIFICATION TRILEMMA

Once exact certification is hard, the direct systems question is not only whether the predicate can be computed, but whether a solver can stay exact, non-abstaining, and polynomial-budgeted on the whole

hard regime. The result below is a direct complexity-theoretic corollary: those three goals are incompatible. We refer to this incompatibility as the *trilemma of exact certification*.

A. Definitions

Definition IX.1 (Certifying Solver). For exact sufficiency on a declared regime Γ , a certifying solver is a pair (Q, V) where:

- Q maps each instance to either ABSTAIN or a candidate verdict with witness,
- V verifies witnesses in polynomial time (in the declared encoding length).

Definition IX.2 (Integrity). A certifying solver is *integrity-preserving* if every non-abstaining output is accepted by V , and every accepted output is correct for the declared exact relation.

Definition IX.3 (Competence). A certifying solver is *competent* on Γ if it is integrity-preserving, non-abstaining on all in-scope instances, and polynomial-budget compliant on all in-scope instances.

Definition IX.4 (Exact Reliability Claim). An *exact reliability claim* on Γ is the conjunction of universal non-abstaining coverage, correctness, and polynomial-budget compliance for exact sufficiency.

Proposition IX.5⁶⁹ (Integrity and Competence Separation). *Integrity and competence are distinct: integrity constrains what can be asserted, while competence adds full coverage under resource bounds.*

Proof. The always-abstain solver is integrity-preserving (it makes no uncertified assertion) but fails competence whenever the in-scope set is non-empty. ■

B. Main Impossibility Theorem

Theorem IX.6⁷⁰ (Integrity-Resource Bound (Conditional)). *Fix a regime Γ whose exact-sufficiency decision problem is coNP-complete under the declared encoding. Under $P \neq \text{coNP}$, no solver is simultaneously integrity-preserving and competent on Γ for exact sufficiency.*

Proof. Assume such a solver (Q, V) exists. We build a polynomial-time decider for exact sufficiency on Γ .

Given an input instance $x \in \Gamma$, run $Q(x)$. Competence implies that Q never abstains on in-scope instances and always halts within polynomial budget. Let y be the returned verdict with witness. Now run the polynomial-time verifier $V(x, y)$.

⁶⁹ : [DQ10](#), [IC33](#) ⁷⁰ : [IC34](#), [DQ31](#)

Integrity implies two things: every non-abstaining output produced by Q is accepted by V , and every accepted output is correct for the declared exact-sufficiency relation. Therefore the combined procedure returns the correct exact verdict on every instance in Γ and runs in polynomial time.

This yields a polynomial-time algorithm for a coNP-complete problem on Γ , hence $\text{coNP} \subseteq \text{P}$. Under $\text{P} \neq \text{coNP}$, this is impossible. Therefore no solver is simultaneously integrity-preserving and competent on Γ . ■

Corollary IX.7⁷¹ (Exact Reliability Impossibility in the Hard Regime). *Under the assumptions of the theorem, exact reliability claims are impossible on the full hard regime.*

Corollary IX.8⁷² (Trilemma of Exact Certification). *In the hard regime, no exact certifier can simultaneously be sound, complete on all in-scope instances, and polynomial-budgeted. Operationally, one of three concessions is unavoidable: abstain on some instances, weaken the guarantee, or risk overclaiming.*

C. Regime-Dependent Abstention

Proposition IX.9⁷³ (Abstention Frontier Expands with Regime Complexity). *For polynomial-time exact certifiers that abstain whenever they cannot certify correctness within budget, the set of instances requiring abstention expands along*

$$\text{static} \rightarrow \text{stochastic} \rightarrow \text{sequential}.$$

Proof. As the decision predicate moves from coNP to PP to PSPACE hardness, exact worst-case certification requires strictly stronger resources. Fixed polynomial resources therefore force abstention on a larger instance family in higher regimes. ■

D. Connection to Simplicity Tax

The trilemma directly relates to the *simplicity tax* (Section X-D). When coordinates relevant for exact decisions are omitted from a central model, the unresolved burden must be handled elsewhere: by local resolution, extra queries, or abstention. In the hard exact regime, Theorem IX.6 implies this burden cannot be discharged for free by a polynomial-budget exact certifier. The principle of *hardness conservation* names this constraint: irreducible certification cost is conserved across the interface, it cannot be eliminated, only displaced.

⁷¹ : DQ28, DQ38 ⁷² : IC30, IC32 ⁷³ : IC32, DQ2

The trilemma thus operationalizes the theoretical impossibility for practical certifiers: exact relevance certification imposes a structural choice among soundness, completeness, and efficiency, with the precise boundary determined by the regime-sensitive complexity map.

X. STRUCTURAL CONSEQUENCES FOR EXACT CERTIFICATION

The regime hierarchy yields both theoretical barriers and practical implications. This section collects structural consequences that follow from the complexity classifications: witness-checking lower bounds, approximation gaps, configuration-simplification limits, regime-dependent certification competence, and the simplicity tax principle. The common theme is that exact relevance certification imposes constraints that cannot be circumvented without either restricting the regime or accepting tradeoffs.

A. Exact Simplification and Over-Specification

Many practical simplification questions are instances of sufficiency checking. For example, configuration parameter reduction asks whether a subset of parameters preserves all decision-relevant behavior, which directly reduces to SUFFICIENCY-CHECK (Appendix A contains detailed reductions). This connection yields immediate complexity consequences.

Corollary X.1⁷⁴ (No General-Purpose Exact Configuration Minimizer). *Assuming $P \neq \text{coNP}$, there is no polynomial-time general-purpose procedure that takes an arbitrary succinctly encoded configuration problem and always returns a smallest behavior-preserving parameter subset.*

Proof. Such a procedure would solve MINIMUM-SUFFICIENT-SET in polynomial time for arbitrary succinctly encoded instances, contradicting Theorem III.5 under the assumption $P \neq \text{coNP}$. ■

Remark X.2⁷⁵. The corollary does not rule out useful simplification tools; it rules out a fully general exact minimizer with worst-case polynomial guarantees. The structured regimes isolated in Section VIII remain viable precisely because they restrict the source of hardness. Moreover, when certification is computationally expensive but parameter maintenance is cheap, over-specification can be cost-optimal—another manifestation of the simplicity tax principle.

B. Regime-Limited Exact Certification

Proposition X.3⁷⁶ (Exact Certification Competence Depends on Regime). *Within the model of this paper, exact certification competence is regime-dependent. In the general succinct regime, exact relevance*

⁷⁴ : DQ36–37 ⁷⁵ : CR1, CT12, DQ26, HD14 ⁷⁶ : DQ36, DQ40, DQ44

certification and exact minimization inherit the hardness results of Sections III and VII. In the structured regimes of Section VIII, exact certification becomes available in polynomial time.

Proof. The negative side is given by Theorems III.4 and III.5, together with the ETH-conditioned lower bounds summarized in Section VII. The positive side is given by the tractability results of Section VIII, which provide explicit polynomial-time certification procedures under structural restrictions. ■

The next three subsections sharpen that basic regime distinction in different directions: exact reliability under polynomial budgets, witness-checking and approximation limits, and the cost of compressing a central interface while leaving exact relevance unresolved.

C. Witness and Approximation Limits

Hardness does not disappear when one weakens exact minimization to witness checking or approximation. The first theorem shows that even the empty-set core can force exponential witness inspection. The remaining results then show two complementary approximation obstructions: a direct gap obstruction on the shifted hard family and an exact optimization transfer on the set-cover gadget family.

1) Witness-Checking Lower Bound:

Definition X.4 (Slot-Inspection Checker). For the empty-set sufficiency core on Boolean state space $S = \{0, 1\}^n$, a *slot-inspection checker* is a deterministic algorithm that adaptively queries witness slots of the form

$$\{(0, z), (1, z)\} \quad z \in \{0, 1\}^{n-1},$$

and, for each queried slot, learns whether the optimizer agrees or disagrees on the two states in that slot. The checker is *sound* if it never accepts a non-sufficient instance and never rejects a sufficient one.

The lower bound below is for deterministic checkers in exactly this access model.

Theorem X.5⁷⁷ (Witness-Checking Lower Bound). *For Boolean decision problems with n coordinates, any sound slot-inspection checker for the empty-set sufficiency core must inspect at least 2^{n-1} witness pairs in the worst case.*

Proof. Let $S = \{0, 1\}^n$. Empty-set sufficiency is exactly the claim that Opt is constant on all states.

Partition S into 2^{n-1} disjoint witness slots

$$\{(0, z), (1, z)\} \quad \text{for } z \in \{0, 1\}^{n-1}.$$

⁷⁷ : WD1-3

Each slot can independently carry a disagreement witness (different optimizer values on its two states).

Consider two instance families:

- *YES instance*: Opt constant on all of S .
- *NO_z instance*: identical to YES except slot z has a disagreement.

If a checker inspects fewer than 2^{n-1} slots, at least one slot z^* is uninspected. On all inspected slots, the YES instance and the tailored NO_{z^{*}} instance are identical. The checker therefore sees exactly the same transcript on these two inputs and must return the same answer on both.

But YES is an empty-set-sufficient instance, whereas NO_{z^{*}} is not. So a common answer cannot be sound on both inputs. Hence any sound worst-case checker must inspect every slot, i.e., at least 2^{n-1} witness pairs. ■

2) *Approximation Gap on the Shifted Family*: For the shifted reduction family used in the mechanization, the optimum support size exhibits a sharp gap:

$$\text{OPT}(\varphi) = 1 \quad \text{if } \varphi \text{ is a tautology,} \quad \text{OPT}(\varphi) = n + 1 \quad \text{if } \varphi \text{ is not.}$$

One coordinate acts as a gate: toggling it changes the optimizer even before the formula variables are considered, so optimum size is never 0. If φ is a tautology, every branch behind that gate behaves identically, so all non-gate coordinates become irrelevant and the singleton gate coordinate is sufficient. If φ is not a tautology, choose a falsifying assignment a . For each formula coordinate i , compare the open reference state with the state that inserts a at coordinate i : these states agree on all other coordinates but induce different optimal-action sets, so coordinate i is relevant. Together with gate relevance, this forces every coordinate to be present in every sufficient set. That is what creates the exact 1 versus $n + 1$ gap. Theorem X.6⁷⁸ (Approximation-hardness theorem on the shifted family). *Fix $\rho \in \mathbb{N}$. Let $\mathcal{A}$ be any solver that, on every shifted-family instance, returns a sufficient set whose cardinality is within factor ρ of optimum. Then for every formula on n coordinates with $\rho < n + 1$,*

$$\varphi \text{ is a tautology} \iff |\mathcal{A}(\varphi)| \leq \rho.$$

Proof. Tautology $\Rightarrow$ threshold passes. If φ is a tautology, the optimum support size on the shifted family is 1, realized by the singleton gate coordinate. A factor- ρ solver therefore returns a sufficient set of size at most $\rho \cdot 1 = \rho$, so $|\mathcal{A}(\varphi)| \leq \rho$.

⁷⁸ : [DQ20](#), [DQ39](#), [DQ43](#)

Threshold passes $\Rightarrow$ tautology. Assume $|\mathcal{A}(\varphi)| \leq \rho$ with $\rho < n + 1$. If φ were not a tautology, then every coordinate would be necessary on the shifted family, so every sufficient set would have size exactly $n + 1$. Since $\mathcal{A}(\varphi)$ is sufficient by assumption on $\mathcal{A}$, this would force $|\mathcal{A}(\varphi)| = n + 1 > \rho$, contradiction.

Therefore $|\mathcal{A}(\varphi)| \leq \rho$ holds exactly in the tautology case. ■

Theorem X.7⁷⁹ (Counted-runtime threshold decider). *Fix $\rho \in \mathbb{N}$. Let $\mathcal{A}$ be any counted polynomial-time factor- ρ solver for the shifted minimum-sufficient-set family. Then the derived procedure*

$$\varphi \mapsto \mathbf{1}\{|\mathcal{A}(\varphi)| \leq \rho\}$$

is a counted polynomial-time tautology decider on the gap regime $\rho < n + 1$.

Proof. Correctness is exactly the gap-threshold theorem above: on the regime $\rho < n + 1$, the predicate $|\mathcal{A}(\varphi)| \leq \rho$ is equivalent to tautology.

For runtime, the derived decider performs two operations: one call to the counted polynomial-time solver $\mathcal{A}$, followed by one cardinality comparison against the fixed threshold ρ . The second step contributes only constant additive overhead relative to the solver run. Hence the derived decision procedure remains counted polynomial-time. ■

3) Set-Cover Transfer Boundary:

Theorem X.8⁸⁰ (Exact set-cover equivalence on the gadget family). *On the mechanized gadget family, a coordinate set is sufficient if and only if the corresponding set family is a cover. In particular, feasible solutions are in bijection and optimum cardinalities coincide exactly.*

Proof. The gadget is constructed so that each universe element u gives two states, (false, u) and (true, u) , with different optimal-action sets. A coordinate i distinguishes this pair exactly when the corresponding set covers u .

If I is not a cover, choose an uncovered universe element u . Then every coordinate in I takes the same value on (false, u) and (true, u) , but their optimal-action sets differ. So I is not sufficient.

If I is a cover, then for every universe element there exists some coordinate in I that separates the two tagged states attached to that element. Hence two states that agree on all coordinates in I cannot disagree in the tag component in a way that changes the optimizer. The optimizer therefore factors through the I -projection, so I is sufficient.

⁷⁹ : DQ23–24 ⁸⁰ : DQ11, DQ34

This yields an exact bijection between feasible sufficient sets and feasible covers, preserving cardinality. Therefore optimum values coincide on the gadget family. ■

Corollary X.9⁸¹ (Conditional approximation transfer). *Any factor guarantee or instance-dependent ratio guarantee for minimum sufficiency on the gadget family induces the same guarantee for set cover on that family. Consequently, any impossibility result proved for set cover on that family transfers immediately to minimum sufficiency.*

Proof. Compose the candidate minimum-sufficiency solver with the gadget translation and then use the exact equivalence theorem above. Because feasible sets correspond bijectively and preserve cardinality, both approximation factors and instance-dependent ratios are unchanged. ■

The two reductions serve different purposes. The shifted family gives a direct gap obstruction: factor approximation already decides tautology there. The set-cover family gives an exact optimization equivalence: sufficiency is cover there. Together they show that weakening exact minimization to witness checking or approximation does not dissolve the core obstruction.

D. Externalized Relevance and Simplicity Tax

The final implication concerns architectural compression. Compression relocates relevance, it does not remove it. When coordinates relevant for exact decisions are omitted from a central model, the unresolved burden must be handled elsewhere in the system: by local resolution, extra queries, or abstention. The *simplicity tax* is the name for that externalized burden.

Corollary X.10⁸² (Hardness Conservation). *Fix a model M and decision problem $\mathcal{D}$. Coordinates in $R(\mathcal{D}) \setminus A_M$ are still decision-relevant for exact behavior preservation. Hence any exact deployment must handle them somewhere: by enlarging the central model, by performing local resolution, by demanding extra queries, or by abstaining on some cases. In the hard exact regime, Theorem IX.6 implies that this burden cannot be discharged for free by a polynomial-budget exact certifier.*

Proof. Take any coordinate $i \in R(\mathcal{D}) \setminus A_M$. Because i is relevant, there exist witness states whose optimal-action sets differ in a way that cannot be ignored in any exact representation. If a deployment neither models i centrally nor resolves its effect elsewhere, then its exact behavior would factor through the smaller interface A_M , making every coordinate outside A_M irrelevant. That contradicts $i \in R(\mathcal{D})$.

⁸¹ : DQ25, DQ32–33 ⁸² : DQ9, HD23

So omitted relevant coordinates must be handled somewhere outside the central interface. In the hard exact regime, Theorem IX.6 shows that polynomial-budget exact competence is unavailable under $P \neq \text{coNP}$. Therefore this external burden cannot, in general, be eliminated for free by a polynomial-budget exact certifier. ■

The simplicity tax operationalizes the certification trilemma (Corollary 7.3) for architectural compression. When exact relevance cannot be certified, systems cannot simply discard coordinates without consequence: the unresolved burden is conserved across the interface and must be paid elsewhere. This principle connects the complexity map to practical deployment constraints: the cost of exact certification is either paid at the center through computational investment, or distributed to the sites as externalized handling.

E. Artifact Scope and the Finite-to-Asymptotic Boundary

The Lean 4 artifact verifies the finite combinatorial core used throughout Sections 3–5: reduction correctness with size bounds, explicit-state deciders and step-counted searches, bridge lemmas, and witness/checking schemas used by stochastic upper-bound arguments. The artifact checks the finite mechanisms—reduction constructions, witness structures, explicit-state deciders—that underpin the paper’s asymptotic complexity claims. The asymptotic class placements (coNP-completeness, PP-hardness, PSPACE-completeness, NP^{PP} membership) are argued in the text using standard complexity-theoretic conventions, as is typical for the field.

The paper-level layer is the asymptotic lift: oracle-machine formulations, class memberships, and ETH-transfer arguments. In particular, the artifact does not formalize oracle Turing machines or polynomial-space machine semantics; those are argued in the manuscript using standard complexity-theoretic conventions.

So the classification is intentionally hybrid: finite gadgets and decision procedures are mechanically checked, while coNP/PP/PSPACE/ NP^{PP} placements and ETH consequences are proved in text.

Summary of what is verified vs. what is assumed.

The finite core is solid; the asymptotic lifts are standard but not mechanized.

Claim	Verified in Lean	Paper-level
Reduction correctness	Polynomial-size output, truth-value preservation	—
Explicit-state decidability	Exact algorithm with step bound	—
Witness/checking pattern	Finite existential/universal structure	Oracle-machine encoding
coNP-completeness	TAUTOLOGY $\rightarrow$ empty-set sufficiency (finite)	Asymptotic class membership
PP-hardness	MAJSAT $\rightarrow$ stochastic decisiveness (finite)	Majority-vote characterization
PSPACE-completeness	TQBF $\rightarrow$ sequential sufficiency (finite)	Polynomial-space TM theory
NP^{PP} upper bound	Witness/checking schema	Oracle machine construction
ETH lower bound	Reduction to empty-set sufficiency	Asymptotic reduction chain

TABLE VI: Summary of what is verified in Lean vs. what is argued at paper-level.

XI. RELATED WORK

A. Formalized Complexity Theory and Mechanized Reductions

Machine verification of complexity-theoretic arguments remains substantially less mature than formal verification in algebra, analysis, or standard algorithmics. Forster et al. [5] developed certified machine models and computability infrastructure in Coq, and Kunze et al. [6] formalized major proof-theoretic components in Coq as well. In Isabelle/HOL, much of the mature work has centered on verified algorithms and resource analysis for concrete procedures rather than on families of hardness reductions [7], [8]. Lean 4 and Mathlib provide increasingly capable foundations for mechanized finite mathematics and computation [9], [10], but reusable reduction suites for $\text{coNP}/\Sigma_2^P/\text{PP}/\text{PSPACE}$ -style arguments remain relatively rare.

The present artifact is intended as a problem-specific certification layer within that broader program. It does not claim to settle formal complexity theory in full generality; instead, it internalizes the reduction-correctness lemmas, size-bounded hardness packages, exact finite deciders, counted-search procedures, tractability wrappers, explicit-state upper-bound interfaces, and the finite witness/checking machinery now used for the stochastic NP^{PP} upper-bound story. In that sense the contribution is closest to a certified reduction-and-decision core for one theorem family rather than to a general-purpose complexity library.

B. Rough Sets, Reducts, and Attribute Reduction

The closest classical neighbor is rough-set theory and its large literature on reducts and attribute reduction [11]–[13]. That literature asks when a smaller attribute set preserves discernibility or decision-

table structure, and it has developed both exact and heuristic methods for reduct computation.

C. Backdoor Tractability in CSPs

A closely related line of work studies backdoor tractability for constraint satisfaction problems [2], [14]. In that framework, a *backdoor* is a minimal variable set whose fixation collapses a CSP to a tractable class (e.g., one admitting a majority polymorphism). Bessiere et al. [2] showed that testing membership in tractable classes characterized by majority or conservative Malt'sev polymorphisms is polynomial-time, but finding a minimal backdoor to such a class is W[2]-hard when the tractable subset is unknown and fixed-parameter tractable only when that subset is given as input. This directly parallels our structural results: coordinate sufficiency in decision problems and backdoor tractability in CSPs both formalize the same intuition, that discovering what matters is fundamentally harder than exploiting a known structure. Our bounded-treewidth reduction (Theorem VIII.13) explicitly connects to their CSP algorithms and backdoor methodology. Subsequent work in parameterized complexity has rigorously bounded the tractability of discovering these minimal structural sets across varying graph widths and backdoor types [15].

The complexity of computing minimal reducts has been studied in the rough-set literature. Finding a minimal attribute set that preserves decision-distinctions is known to be NP-hard in general [16], and more refined analysis places certain reduct variants in Σ_2^P [17]. These results establish that the static regime questions studied here, whether a coordinate set preserves decision distinctions, and whether a minimal such set exists, fall within a known complexity landscape for attribute reduction.

The key structural difference is this: general decision tables can admit multiple incomparable minimal reducts, necessitating a combinatorial search through the reduct lattice. Under the exact optimal-action preservation predicate studied here, Proposition II.9 proves there is exactly one minimal sufficient set: the relevant-coordinate set $R(\mathcal{D})$ itself. This uniqueness follows from the optimizer quotient structure: states in the same quotient class share the same optimal-action set, so any sufficient set must separate all quotient classes, and the relevant-coordinate set is precisely the minimal set achieving this separation.

The complexity consequence is significant. In general rough-set reduct computation, multiple minimal reducts can exist, placing minimal-reduct search in Σ_2^P . Here, uniqueness collapses the search problem: finding a minimal sufficient set reduces to checking which coordinates are relevant (coNP-complete). The Σ_2^P search layer vanishes because there is no need to search among incomparable candidates; the single minimal set is uniquely determined by relevance. This structural collapse—from Σ_2^P search to coNP

relevance containment—is driven by optimizer-quotient uniqueness and is, to our knowledge, a novel classification not captured by general rough-set bounds.

Both settings ask which coordinates can be removed without losing decision distinctions.

Proposition XI.1⁸³ (Static Sufficiency as Reduct Preservation for the Induced Decision Table). *Given a finite decision problem $\mathcal{D} = (A, X_1, \dots, X_n, U)$, form the induced decision table whose condition attributes are the coordinates X_i and whose decision label at state s is the set-valued class label $\text{Opt}(s) \subseteq A$. Then a coordinate set I is sufficient in the sense of this paper if and only if the projection to I preserves the decision label of the induced table. Consequently, the minimal sufficient sets of $\mathcal{D}$ are exactly the reducts of this induced decision table.*

Proof. By definition, I is sufficient exactly when any two states agreeing on I have the same optimal-action set. In the induced decision table, the decision label of state s is precisely $\text{Opt}(s)$, so the same condition says that projection to I preserves the decision label. Minimality on the present side is therefore exactly reduct minimality for the induced table. ■

Rough-set reducts are formulated relative to indiscernibility relations or decision tables, whereas the object here is the optimizer map of a decision problem and the induced exact preservation of optimal-action sets. Proposition XI.1 shows that the static regime can be recast as reduct preservation for the induced decision table. What the present paper adds is: (i) the collapse theorem showing minimum sufficiency collapses to relevance containment, bypassing the general NP-hardness barrier for exact minimization; and (ii) the systematic extension to stochastic and sequential regimes, which takes the reduct question beyond the classical rough-set setting. The stochastic and sequential layers introduce conditional-comparison and temporal-structure phenomena that are outside the usual rough-set formulation.

The same generosity is due to the broader feature-selection and subset-selection literature in machine learning and combinatorial optimization [18], [19]. Those works study sparse predictive representations, informative feature subsets, and the computational cost of choosing them. Our setting differs in targeting exact preservation of the optimal-action correspondence rather than prediction loss, classifier complexity, or approximate empirical utility. Still, the shared concern is clear: identifying what information matters and what can be safely omitted.

⁸³ : DP1–2

D. Decision-Theoretic Sufficiency, Informativeness, and Value of Information

Classical decision theory and informativeness ask when one information structure is at least as useful as another [20]–[23]. Our setting is the computational certification layer of that question: exact coordinate projection preserving optimal-action correspondence.

E. Abstraction, Bisimulation, and Homomorphisms in Planning and Reinforcement Learning

The stochastic and sequential parts of the paper sit near a second major literature: exact abstraction in Markov decision processes, POMDPs, bisimulation-based aggregation, and MDP homomorphisms [24]–[28]. This body of work studies when state spaces can be aggregated while preserving value or policy structure, and it has established many of the representation-sensitive complexity phenomena that make richer stochastic and sequential models computationally difficult.

That literature is structurally close to the present paper and should be credited as such. The main difference is that our predicate is coordinate sufficiency for preserving optimal-action sets under an explicit coordinate-hiding operation. We do not study arbitrary abstract state maps, approximate value-function guarantees, or the quantitative bisimulation metrics pioneered by Ferns et al. [29] to measure behavioral similarity. While recent work has successfully leveraged such metrics to learn robust, task-invariant representations in high-dimensional deep reinforcement learning (e.g., Zhang et al. [30]), our focus remains strictly on the combinatorial complexity of exact certification. Instead of optimizing a continuous representation space to discard task-irrelevant distractors, we ask whether one can exactly certify that no omitted coordinate changes the optimal-action correspondence. The contribution here is therefore not abstraction theory in general, but a unified exact-certification account in which static, stochastic, and sequential regimes are analyzed as variants of one decision-relevance question.

F. Explanatory and Structural Reasoning Links

Anchor-style explanations, abductive explanations, and exact reason/justification frameworks all study small structures sufficient for a decision outcome [31]–[34]. Our contribution is not a new explanation semantics but a regime-sensitive complexity classification for exact coordinate-preservation predicates.

The same applies to causality/responsibility traditions [35]–[38]: they motivate relevance notions, while this paper studies the algorithmic certification cost of exact preservation under coordinate hiding.

G. Oracle, Query, and Access-Based Lower Bounds

Query-access lower bounds provide another nearby technique family [39]. Our witness-checking duality and access-obstruction results belong to that tradition in spirit. The difference is that the lower bounds are developed around the same fixed sufficiency predicate used throughout the paper, which allows direct comparison between forward evaluation, certification, exact minimization, and restricted-access models without changing the underlying decision relation.

H. Scope and Novelty

The paper is intentionally synthetic. Rough sets and reducts study decision-preserving attribute reduction; feature selection studies informative subsets; statistical decision theory studies informativeness and sufficiency; abstraction and bisimulation literatures study exact aggregation of stochastic and sequential decision processes; explanation methods study local decision-preserving cores; formalized complexity work studies certified reductions and machine-checked decision procedures. The contribution here is to treat exact relevance certification itself as the common object and to develop one coherent complexity-theoretic account around it.

What is distinctive in the present paper is not an isolated static or sequential hardness theorem taken on its own. The static regime can be recast as reduct preservation for the induced decision table, and the sequential row belongs near existing exact abstraction phenomena in richer control models. The contribution here is to formalize one exact decision-preservation predicate in optimizer language, track it coherently across static, stochastic, and sequential regimes, expose the preservation/decisiveness split in the stochastic case, identify the tractable and intractable boundaries of exact certification, and support the resulting finite combinatorial core with mechanized verification. The paper’s novelty is therefore best understood as a unified exact-certification theory rather than as a single isolated theorem.

XII. CONCLUSION

This paper develops a regime-sensitive theory of exact relevance certification for coordinate-structured decision problems. Its aim is to characterize what information a decision actually depends on, how that information is represented by the optimizer quotient, and when exact certification of that dependence is tractable, split, or fundamentally limited across static, stochastic, and sequential regimes.

Main Results

The core classification is Theorem VI.1, with cross-encoding separation in Theorem VII.1, reliability limits in Theorem IX.6, and structured tractability in Theorem VIII.1. Together, these results show that exact relevance certification is computationally distinct from pointwise optimizer evaluation and depends sharply on regime (Theorem VI.1). The optimizer quotient remains the governing structure (Theorem II.15), and the remaining conjecture and open problem mark the boundary left unresolved.

Implications

The implications are direct hardness transfers. Configuration simplification is an instance of sufficiency checking, so exact behavior-preserving minimization has no general-purpose polynomial-time worst-case solution unless $P = \text{coNP}$. The same issue appears in real pipeline orchestration settings, where a nested configuration object controls well selection, output layout, and downstream materialization. There too the practical question is whether one can hide part of the configuration interface without changing the induced compilation or execution decisions. Our results show that this is an exact certification problem whose worst-case difficulty depends on whether one is in the static, stochastic, or sequential regime. Reliable exact certification is likewise regime-limited: under $P \neq \text{coNP}$, the trilemma of exact certification rules out any solver that is simultaneously sound, complete on the full hard regime, and polynomial-budgeted. And once the relevant-coordinate set $R(\mathcal{D})$ is fixed, any compressed interface merely partitions that relevance into centrally handled and externalized parts. Compression therefore relocates exact relevance rather than erasing it; under the per-site cost model, the unresolved part grows linearly with deployment scale.

This is also the practical meaning of the simplicity tax. In the hard exact regime, a cheap simplification procedure cannot be assumed to have removed decision-relevant structure merely because it has compressed the visible interface. Any relevance omitted from the certified core must still be paid for somewhere else in the system: by local resolution, extra queries, abstention, or weaker guarantees. Exact simplification is therefore costly, but inexact simplification is not free; it externalizes the unresolved burden.

Mechanization

The proof artifact mechanically checks the main reduction constructions, hardness packages, finite deciders, search procedures, tractability statements, step-counting wrappers, the stochastic sufficiency bridge

package, the explicit-state decision procedures for the preservation family, the full-support inheritance results, the new support-sensitive obstruction and bridge lemmas for preservation, the decisiveness upper bounds, and the finite witness/checking schemata for the stochastic existential classification⁸⁴ Full oracle-machine formalization in Lean remains outside the current state of formalized complexity theory; the paper proves the oracle-class memberships in the text following standard conventions, while the artifact provides independent verification of the finite combinatorial core. This places the mechanization in the emerging category of problem-specific certified reduction infrastructure.

Outlook

Two immediate directions remain. First, the reduction infrastructure can be integrated more tightly with general-purpose formalized complexity libraries. Second, the artifact boundary can be made even cleaner by continuing to package the existing finite-decision results into more uniform summary interfaces without changing the paper’s complexity claims. On the complexity side, the remaining gap is now sharply localized: stochastic decisiveness already has succinct hardness and a verified witness/checking core for the oracle-class memberships proved in the paper text, but a fully standard oracle-machine formalization of the resulting oracle-class placement and of the NP^{PP} boundary for the anchor and minimum decisiveness queries remains outside the current repository; on the preservation side, Conjecture [IV-D0a](#) and Open Problem [IV-D0a](#) sharpen the remaining open terrain to the full general-distribution and succinct-encoding complexity of the minimum and anchor preservation variants. The mechanized artifact already exposes the finite combinatorial core and support-sensitive lemmas that isolate this obstruction; see the supplementary material for representative Lean handles when framing the open problem.⁸⁵

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⁸⁴ : [DC91–99](#), [OU8–12](#), [EH2–10](#).

⁸⁵ : [OU12](#), [DC96](#), [DC98](#)

The author retained full intellectual and editorial control, including problem selection, theorem statements, assumptions, novelty framing, acceptance criteria, and final inclusion/exclusion decisions. No technical claim was accepted solely from AI output. Formal claims reported as machine-verified were admitted only after Lean verification (no `sorry` in cited modules) and direct author review; Lean was used as an integrity gate for responsible AI-assisted research. The author is solely responsible for all statements, citations, and conclusions.

APPENDIX

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APPENDIX

The full Lean-handle ledger and claim-to-handle mapping are provided in the supplementary PDF (and archived artifact at <https://doi.org/10.5281/zenodo.19057595>). We omit those long tables from the main manuscript body.

Mechanized Witness Schemas and Oracle-Class Arguments

The paper uses standard oracle-class language for several stochastic upper bounds, especially the conP^{PP} -style complement packaging for decisiveness and the NP^{PP} upper bounds for the anchor and minimum decisiveness queries. The corresponding oracle-class memberships are proved in the paper text, following standard complexity-theoretic proof conventions. The proof artifact complements those proofs by mechanizing the finite witness/checking schemata and reduction cores that the oracle arguments require.

Concretely, the mechanized layer certifies ingredients of the following form: a bad fiber witnesses failure of decisiveness; an existentially guessed coordinate set or anchor witness reduces the anchor/minimum queries to the corresponding decisiveness-style verifier; and the bounded explicit-state procedures agree with the abstract predicates they are intended to decide. The paper then translates these certified witness/checking packages into the stated oracle-class membership arguments by the usual guess-and-query reasoning in complexity theory. Thus the artifact provides independent verification of the finite combinatorial core, while the oracle-class interpretation is established in the manuscript.

This appendix collects applied translations that instantiate the main predicates from the core theory.

A. Configuration Case Study

Consider a simplified service configuration with three parameters: cache mode $p_1 \in \{\text{off}, \text{on}\}$, retry policy $p_2 \in \{1, 3\}$, and replica count $p_3 \in \{1, 2\}$. Let the observable behavior set be $B = \{\text{latency-ok}, \text{write-safe}, \text{cost-ok}\}$. Suppose enabling the cache affects only latency, increasing retries affects only write safety, and replica count affects write safety and cost. By the sufficiency characterization (Proposition II.18), the simplification question reduces to an exact sufficiency question on the induced decision problem: which parameter subsets preserve the optimal-behavior set? In this toy instance, if the target simplification must preserve write safety and cost but is indifferent to latency, the exact behavior-preserving core is $\{p_2, p_3\}$, while p_1 is irrelevant to that target behavior set.⁸⁶

B. Toy POMDP Translation

Definition A.1 (One-step POMDP). A one-step POMDP is a tuple (S, A, O, p, o, r) where S, A, O are finite sets of states, actions, and observations, $p \in \Delta(S)$ is a prior distribution, $o : S \rightarrow O$ is the observation map, and $r : A \times S \rightarrow \mathbb{R}$ is the immediate reward function.

Write $\text{Opt}_{\text{full}}(s) := \arg \max_{a \in A} r(a, s)$ for the full-information optimal-action set and, for an observation value $o \in O$, set

$$\text{Opt}_{\text{obs}}(o) := \arg \max_{a \in A} \mathbb{E}_{S \sim p}[r(a, S) \mid o(S) = o].$$

Let $g : O \rightarrow O'$ be any coarsening of observations and write $\phi := g \circ o : S \rightarrow O'$.

Proposition A.2⁸⁷ (Reduction to stochastic preservation). *Let (S, A, O, p, o, r) be a one-step POMDP and let $\phi : S \rightarrow O'$ be as above. Define the decision problem $\mathcal{D} = (A, S, U)$ with $U(a, s) := r(a, s)$ and state distribution p . Then ϕ is stochastically preserving for $\mathcal{D}$ in the sense of Section IV if and only if*

$$\forall s \in S : \quad \text{Opt}_{\text{full}}(s) = \text{Opt}_{\text{obs}}(\phi(s)).$$

Proof. By construction $U(a, s) = r(a, s)$ so the full-information optimizer in $\mathcal{D}$ equals $\text{Opt}_{\text{full}}(s)$. The coarse-induced optimizer is, by definition, the set maximizing conditional expectation given ϕ , which is exactly $\text{Opt}_{\text{obs}}(\phi(s))$. ■

a) Worked instance.: Let $S = \{s_1, s_2\}$, $A = \{a, b\}$, $p(s_1) = p(s_2) = 1/2$, and

$$r(a, s_1) = 2, \quad r(b, s_1) = 1, \quad r(a, s_2) = 0, \quad r(b, s_2) = 3.$$

⁸⁶ : CR1, CT12.

⁸⁷ : EX1

If observations are fully coarsened to one symbol, then

$$\mathbb{E}[r(a, S) \mid \phi] = 1, \quad \mathbb{E}[r(b, S) \mid \phi] = 2,$$

so the coarse optimizer is $\{b\}$ while $\text{Opt}_{\text{full}}(s_1) = \{a\}$. Hence preservation fails.⁸⁸

C. Hyperparameter-Redundancy Translation

Fix finite environment set E and finite domains $X_\alpha, X_\gamma, X_\epsilon$. Let

$$H := X_\alpha \times X_\gamma \times X_\epsilon,$$

and for each $e \in E$, $f_e : H \rightarrow \mathbb{R}$ be expected return. Define

$$\mathcal{D}_{\text{hp}} = (A, H, U), \quad A := E, \quad U(e, h) := f_e(h).$$

Let $\pi : H \rightarrow H' := X_\alpha \times X_\epsilon$ drop coordinate γ .

Proposition A.3⁸⁹ (Reduction to a static preservation check). *With notation as above, “ γ is redundant for tuning” is equivalent to static preservation for $\mathcal{D}_{\text{hp}}$ under projection π . Concretely, γ is redundant iff*

$$\forall e \in E \forall h \in H : \quad h \in \text{Opt}_e \iff \exists h_0 \in \text{Opt}_e \text{ with } \pi(h_0) = \pi(h),$$

where $\text{Opt}_e := \arg \max_{h \in H} f_e(h)$.

Proof. By definition, redundancy means full-space maximizers and projected maximizers correspond through π . Rewriting this correspondence as equality of induced optimal-action sets gives exactly the static preservation predicate for the derived decision problem. ■

REFERENCES

- [1] R. Impagliazzo and R. Paturi, “On the complexity of k -sat,” *Journal of Computer and System Sciences*, vol. 62, no. 2, pp. 367–375, 2001.
- [2] C. Bessiere, C. Carbonnel, E. Hebrard, G. Katsirelos, and T. Walsh, “Detecting and exploiting subproblem tractability,” in *Proceedings of the Twenty-Third International Joint Conference on Artificial Intelligence (IJCAI)*, 2013, pp. 468–474.
- [3] H. L. Bodlaender, “A tourist guide through treewidth,” in *Acta Cybernetica*, vol. 11, 1993, pp. 1–21.
- [4] E. C. Freuder, “Complexity of k -tree structured constraint satisfaction problems,” *Proceedings of AAAI*, pp. 4–9, 1990.

⁸⁸ : EX3.

⁸⁹ : EX2

- [5] Y. Forster, E. Heiter, and G. Smolka, “Verified programming of Turing machines in Coq,” in *Proceedings of the 8th ACM SIGPLAN International Conference on Certified Programs and Proofs*. ACM, 2019, pp. 114–127.
- [6] F. Kunze, G. Smolka, and Y. Forster, “Formal small-step verification of a call-by-value lambda calculus machine,” in *Asian Symposium on Programming Languages and Systems*. Springer, 2019, pp. 264–283.
- [7] T. Nipkow, L. C. Paulson, and M. Wenzel, *Isabelle/HOL: A Proof Assistant for Higher-Order Logic*. Springer, 2002.
- [8] M. P. L. Haslbeck and T. Nipkow, “Verified analysis of list update algorithms,” in *Journal of Automated Reasoning*, vol. 65. Springer, 2021, pp. 117–156.
- [9] The mathlib Community, “The Lean mathematical library,” *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pp. 367–381, 2020.
- [10] L. de Moura and S. Ullrich, “The lean 4 theorem prover and programming language,” in *International Conference on Automated Deduction*. Springer, 2021, pp. 625–635.
- [11] Z. Pawlak, “Rough sets,” *International Journal of Parallel Programming*, vol. 11, no. 5, pp. 341–356, 1982.
- [12] R. Jensen and Q. Shen, “Semantics-preserving dimensionality reduction: Rough and fuzzy-rough-based approaches,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 16, no. 12, pp. 1457–1471, 2004.
- [13] R. Jensen, “Rough set-based feature selection: A review,” in *Rough Computing: Theories, Technologies and Applications*. IGI Global, 2008, pp. 91–129.
- [14] R. Williams, C. P. Gomes, and B. Selman, “Backdoors to typical case complexity,” in *Proceedings of the Eighteenth International Joint Conference on Artificial Intelligence (IJCAI)*. Morgan Kaufmann Publishers Inc., 2003, pp. 1173–1178.
- [15] R. Ganian, M. S. Ramanujan, and S. Szeider, “Discovering archipelagos of tractability for constraint satisfaction and counting,” in *Proceedings of the 27th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. SIAM, 2016, pp. 1670–1681.
- [16] R. Slowinski and D. Vanderpooten, “Decision under uncertainty: A rough set approach,” *European Journal of Operational Research*, vol. 80, no. 2, pp. 277–289, 1995.
- [17] X. Zhang *et al.*, “On the complexity of attribute reduction in decision tables,” *Fundamenta Informaticae*, vol. 123, no. 4, pp. 421–434, 2013.
- [18] A. L. Blum and P. Langley, “Selection of relevant features and examples in machine learning,” *Artificial Intelligence*, vol. 97, no. 1-2, pp. 245–271, 1997.
- [19] E. Amaldi and V. Kann, “On the approximability of minimizing nonzero variables or unsatisfied relations in linear systems,” *Theoretical Computer Science*, vol. 209, no. 1-2, pp. 237–260, 1998.
- [20] D. Blackwell, “Equivalent comparisons of experiments,” *The Annals of Mathematical Statistics*, vol. 24, no. 2, pp. 265–272, 1953.
- [21] L. J. Savage, *The Foundations of Statistics*. John Wiley & Sons, 1954.
- [22] H. Raiffa and R. Schlaifer, *Applied Statistical Decision Theory*. Harvard University Press, 1961.
- [23] R. A. Howard, “Information value theory,” *IEEE Transactions on Systems Science and Cybernetics*, vol. 2, no. 1, pp. 22–26, 1966.
- [24] C. H. Papadimitriou and J. N. Tsitsiklis, “The complexity of markov decision processes,” *Mathematics of Operations Research*, vol. 12, no. 3, pp. 441–450, 1987.
- [25] M. L. Littman, J. Goldsmith, and M. Mundhenk, “The computational complexity of probabilistic planning,” *Journal of Artificial Intelligence Research*, vol. 9, pp. 1–36, 1998.
- [26] M. Mundhenk, J. Goldsmith, C. Lusena, and E. Allender, “The complexity of finite-horizon markov decision process problems,” *Journal of the ACM*, vol. 47, no. 4, pp. 681–720, 2000.
- [27] R. Givan, T. Dean, and M. Greig, “Equivalence notions and model minimization in markov decision processes,” *Artificial Intelligence*, vol. 147, no. 1-2, pp. 163–223, 2003.

- [28] B. Ravindran, “An algebraic approach to abstraction in reinforcement learning,” Ph.D. dissertation, University of Massachusetts Amherst, 2004.
- [29] N. Ferns, P. Panangaden, and D. Precup, “Metrics for finite markov decision processes,” *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence (UAI)*, pp. 162–169, 2004.
- [30] A. Zhang, R. McAllister, R. Calandra, Y. Gal, and S. Levine, “Learning invariant representations for reinforcement learning without reconstruction,” *International Conference on Learning Representations (ICLR)*, 2021.
- [31] M. T. Ribeiro, S. Singh, and C. Guestrin, “Anchors: High-precision model-agnostic explanations,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, 2018.
- [32] A. Ignatiev, N. Narodytska, and J. Marques-Silva, “Abduction-based explanations for machine learning models,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, pp. 1511–1519, 2019.
- [33] J. Marques-Silva and A. Ignatiev, “Delivering trustworthy ai through formal xai,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, pp. 12 342–12 350, 2022.
- [34] A. Darwiche and A. Hirth, “On the (complete) reasons behind decisions,” *Journal of Logic, Language and Information*, vol. 32, no. 1, pp. 63–88, 2023.
- [35] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press, 2009.
- [36] P. Spirtes, C. Glymour, and R. Scheines, *Causation, Prediction, and Search*, 2nd ed. MIT Press, 2000.
- [37] J. Y. Halpern and J. Pearl, “Causes and explanations: A structural-model approach. part ii: Explanations,” in *Proceedings of the 17th International Joint Conference on Artificial Intelligence*, 2001, pp. 27–34.
- [38] H. Chockler and J. Y. Halpern, “Responsibility and blame: A structural-model approach,” *Journal of Artificial Intelligence Research*, vol. 22, pp. 93–115, 2004.
- [39] S. Dobzinski and J. Vondrák, “From query complexity to computational complexity,” in *Proceedings of the 44th ACM Symposium on Theory of Computing*, 2012, pp. 1107–1116.